

Applying mixed-integer simulation optimization for tactical design decisions of robotic sorting system with guaranteed security level to combat illicit trade[☆]

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ABSTRACT

With the increasing frequency of online e-commerce and free trade, opportunities and amounts for illicit trade have significantly grown, causing serious harm to the economies, societies, environments, and politics of various countries. Due to outdated and obsolete parcel sorting procedures and equipment, logistics providers handling international import parcels, particularly postal service systems, have become vulnerable entry points for illicit goods. Therefore, this study aims to explore and develop tactical design decisions for a robotic sorting system (RSS) with a guaranteed security level to effectively combat illicit trade and goods entering through international import operations. A mixed-integer stochastic optimization model is formulated to derive optimal tactical solutions, including the number of autonomous mobile robots (AMRs) and inspection personnel, the allocation of drop-off points, and the screening probabilities of inspection stations, to minimize system costs while meeting a specified security threshold. Since the objective function of this model is not analytically available, a high-resolution RSS stochastic simulation model is first constructed to estimate this performance measure. Then, a novel mixed-integer simulation optimization algorithm, namely a combination of optimal computing budget allocation-based ranking and selection procedure and adaptive particle and hyperball search (ORS-APHS), is developed to speed up convergence near-optimal tactical design decisions under a limited simulation budget. We collaborate with Chunghwa Post Company in Taiwan to analyze algorithm efficiency and identify key factors influencing total cost and tactical design decisions to combat illicit trade according to their newly implemented international parcel sorting system. Computational experiments demonstrate that the proposed ORS-APHS algorithm outperforms two common existing methods (Genetic Algorithm and Tabu Search) in terms of effectiveness and efficiency. The sensitivity analysis results indicate that changes in the predetermined system security levels, adjustments in the number of available AMRs, variations in physical re-inspection times, and the distribution of drop-off points for arriving parcels all have significant impacts on total system costs and tactical design solutions. To enhance the security of the RSS system in combating illicit trade, the predetermined system security level is particularly a critical factor in determining the screening probabilities of inspection stations. Decision-makers can leverage the managerial insights and guidelines derived from sensitivity analysis to formulate optimal tactical design plans for the RSS system with a guaranteed security level in the future.

1. Introduction

Over the past few decades, global trade has experienced unprecedented growth, encompassing various countries and continents. With the frequent transactions of online e-commerce and the promotion of free trade, the circulation of international parcels across countries has

become very frequent [50]. This has not only caused a significant increase in the number of imported international parcels in each country but has also easily outpaced the existing governance mechanisms, leading to a surge in opportunities for illicit trade. Illicit trade refers to the exchange of goods conducted in violation of local laws, often involving unethical means to obtain money, goods, or other forms of

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value, causing harm to the economy, society, environment, or politics [66,40]. The scope of illicit goods has significantly expanded, now including counterfeit products, drugs, illegal narcotics, wildlife trafficking, human trafficking, and illegal weapons. These illicit goods can easily enter countries during cross-border transportation, causing harm and negative impacts on the social economy, national security, public health, environment, and politics, which has drawn significant global attention. [43,23,3,40,6,36]. ICAIE [36] points out that the illicit money laundered annually from illegal trade amounts to as much as 5 % of the global gross domestic product (GDP), approximately \$3 to \$5 trillion USD. Therefore, effectively combating illicit trade and preventing the import of illicit goods from their supply chains into each country has become a major issue and challenge for almost every government worldwide [3].

An illicit supply chain refers to a network of individuals, organizations, or actors engaged in unlawful activities related to the sourcing, production, logistics, or distribution of illegal goods or services [7,3]. In the past, most governments primarily sought to prevent illicit trade by disrupting these supply chain networks, using methods such as node disruption (targeting organizations or actors), link disruption (interrupting activities), and other similar approaches [38,3,1]. This strategy includes invasive disruption techniques such as proactive interceptions and crackdowns by law enforcement agencies using network surveillance or data analysis to detect and combat members of illicit supply chains and their activities [3], as well as noninvasive approaches such as the formulation of relevant laws and policies and raising consumer awareness to deter people from engaging in illicit activities [18]. However, few studies in past literature have addressed how to improve or design secure logistics or warehousing systems capable of effectively detecting illicit goods during their physical transportation within the supply chain to prevent them from entering the country [8,9,10]. Particularly in postal service systems, the logistics systems handling international import parcels within the global supply chain have become vulnerable entry points for illicit goods due to outdated and antiquated mail/parcel sorting and processing procedures and equipment. [65,58]. To address this issue, this study focuses on how postal service systems or international parcel logistics companies can redesign parcel sorting systems with guaranteed security levels by incorporating emerging logistics and inspection technologies, thereby effectively detecting illicit parcels and combating illicit trade.

Recently, an emerging logistics technology—robotic sorting systems (RSS) equipped with parcel-sorting autonomous mobile robots (AMRs), such as LiBiao's "Mini Yellow" series (<https://www.libiaorobot.com/>) and Tompkins Robotics' tSort (<https://tompkinsrobotics.com/>), has been widely used in e-commerce and postal logistics to improve efficiency and reduce labor costs [8], but it has not addressed the security risks posed by illicit parcels. Similar to robotic mobile fulfillment systems [44], the design and planning decisions involved in implementing typical RSS systems for parcel sorting operations can be divided into strategic, tactical, and operational levels. The strategic design decisions, determined at the initial establishment of the RSS system, encompass layout types, robot path directions, the size of the sorting area, and the location and number of picking stations. Tactical design decisions, which are adjusted periodically (e.g., monthly or quarterly), aim to determine the optimal allocation of critical resources, including the number of AMRs, the number of operators, and the workload assigned to each picking station etc., and significantly impact system performance. Operational planning involves real-time decisions, including AMR scheduling, AMR path planning, and parcel assignment to picking stations etc. However, these typical planning decisions for the RSS systems in the past literature [28,35,52,69] are all determined from the perspective of optimizing system efficiency or minimizing system costs, completely neglecting system security design issues. This system security refers to RSS having a high-level screening mechanism capable of accurately detecting illicit parcels or mail during sorting operations, such as those containing counterfeit goods, illegal drugs, or items related

to wildlife trafficking. As a result, this study seeks to develop a robotic sorting system with guaranteed security level, capable of simultaneously determining the most effective security screening mechanism and the optimal allocation of critical resources at the tactical design level to replace traditional RSS systems. This system not only optimizes operational efficiency but also effectively detects and prevents the intrusion of illicit parcels, ensuring that a specified level of security is maintained.

Based on the aforementioned motivations, this paper focuses on the tactical design decisions for a robotic sorting system with a guaranteed security level, which includes multi-stage security screening processes to effectively intercept potentially illicit international parcels resulting from illegal trade. We first construct a mixed-integer stochastic optimization model to derive the optimal tactical design solutions, including the number of required AMRs, the number of inspection personnel, the allocation of drop-off points at each picking station, and the security screening probability settings for each inspection stage in the system. This stochastic model ensures that the system achieves the specified security level while minimizing the overall expected system costs. Since the objective function of this model is not analytically available and influenced by the complex stochastic nature of RSS system with a guaranteed security level, it is necessary to develop a simulation optimization algorithm to find optimal solutions and establish a high-resolution RSS stochastic simulation model to evaluate the performance of the solutions generated by the algorithm.

Past research has proposed various specifically designed algorithms, but most have focused on addressing simulation optimization problems involving only continuous or integer decision variables [57,30,2,17,29]. Few studies have focused on the development of mixed-integer simulation optimization algorithms, but these algorithms often require the assumption of a "smoothness" structure [63,41,64,60], making them unsuitable for problems lacking performance function gradients, such as the tactical design decisions of the RSS system. Metaheuristic algorithms, such as Genetic Algorithm (GA) and Tabu Search (TS), are widely applied to large-scale simulation optimization problems but have notable drawbacks [2,26,17]. These include being applicable only to single-type variables (e.g., GA for discrete variables), lacking problem-specific search mechanisms, and wasting simulation resources due to excessive replication evaluations.

Therefore, we propose a novel and efficient mixed-integer simulation optimization algorithm named ORS-APHS, which can effectively find the optimal tactical design solutions based on a high-resolution RSS stochastic simulation model under limited simulation budgets. This algorithm consists of two components: (1) an OCBA-based ranking and selection procedure (ORS) to handle integer variables, including the number of required AMRs, the number of inspection personnel, and drop-off point allocation; and (2) adaptive particle and hyperball search (APHS) to handle continuous variables related to system security probability settings at different inspection stages. In addition, ORS-APHS offers three key advantages: (1) generating high-quality initial solutions tailored to the problem structure, (2) integrating global exploration and local exploitation to create promising and high-performing neighboring solutions, and (3) efficiently allocating simulation replications to evaluate and screen for high-quality feasible solutions, thereby avoiding the waste of simulation resources on inferior solutions. By collaborating with Chunghwa Post in Taiwan, an empirical study based on the newly designed international import parcel sorting system is conducted to evaluate the performance efficiency of the ORS-APHS algorithm and identify key factors influencing total cost and tactical design decisions to combat illicit trade. Computational experiments demonstrate that the proposed ORS-APHS algorithm outperforms two common existing methods (Genetic Algorithm and Tabu Search) in terms of the computational efficiency and solution quality. Sensitivity analysis is performed to investigate the impact of changes in system security levels, variations in the number of available AMRs, adjustments in physical re-inspection times, and distributions of different drop-off points for arriving parcels on total system costs and tactical design

solutions, providing decision-makers with valuable managerial implications for system design.

The remainder of this paper is organized as follows. Section 2 provides a literature review and identifies the research gaps and distinctions between previous studies and this paper. Section 3 defines the tactical design problem for robotic sorting systems with guaranteed security levels and formulates this problem as a mixed-integer stochastic optimization model. Section 4 proposes an efficient mixed integer simulation optimization algorithm called ORS-APHS to handle the tactical design problem for robotic sorting systems with guaranteed security levels. Section 5 investigates a case study of the redesign of the international import parcel sorting system in the new mail processing center in Taiwan as well as the computational results and managerial implications of this empirical study. Finally, Section 6 presents our conclusions and outlines potential avenues for future research.

2. Literature review

2.1. Illicit trade and supply chain

Due to the growth of online e-commerce and the promotion of international free trade, how to prevent illicit trade and the entry of illegal goods (e.g. drugs, counterfeit, arms, wildlife) has become a significant issue for almost every government in the world [3,65]. However, past literature in recent years has mostly focused on preventing the spread of illicit trade from the view of supply chain network model [37,47,3,32,48,4,1,31,27]. Jabarzare et al. [37] presented a mixed-integer programming model by reformulating a Min–Max bi-level mathematical approach to investigate the disruption of illegal product distribution in a network comprising multiple sources and destinations. Mirzaei et al. [47] introduced two bi-level, two-stage mixed-integer models aimed at disrupting the smuggling of goods within a supply chain network, addressing both continuous and discrete interdiction scenarios. Najafi et al. [48] developed a bi-level mixed-integer programming model and Benders decomposition algorithm to manage resources in an illicit smuggling network with uncertain flows and unidentified structure, aiming to minimize illicit flows and interdict maximum flow in a multi-commodity, multi-source network. Anzoom et al. [4] proposed a graph-theoretic approach to analyze the composition of illicit supply networks under limited information by constructing a comprehensive network of entities suspected of participating in the illicit supply chain and identifying a set of dissimilar supply chain structures that criminals might adopt. Akhavantaheri et al. [1] conducted an in-depth analysis of various policies within the electronics industry and their potential effects on the illicit supply chain of counterfeit electronics. Griffin et al. [31] explored a network interdiction problem targeting wildlife trafficking and propose a novel model to address the critical challenges in Illicit Wildlife Trade (IWT). Huang et al. [27] first analyzed seizure data, shipping records, and investigation reports to gain a deeper understanding of the illicit supply chains for counterfeit N95 respirators. They then developed a bi-level multi-period optimization model that accounts for the behaviors of both counterfeiters and disruptors, evaluating the effectiveness of five disruption strategies in combating counterfeit respirator supply chains: website shutdowns, wholesale seizures, enforcement actions against shell companies, warehouse seizures, and customs interceptions.

Due to the complexity of illicit supply chains, a few emerging studies have begun utilizing discrete-event simulation to model and explore their dynamic operations and relationships [39,33,61,42,62], further integrating simulation optimization algorithms to develop optimal disruption strategies and interception decisions for illicit goods [33,42]. Kersten et al. [39] proposed a discrete-event simulation framework for illicit goods detection in large-volume freight, which considers logistical conditions like available space and container flow patterns to help customs analyze and compare the impact of various detection architectures on the efficiency of goods flow through security checkpoints,

such as average inspection time. Hashemi et al. [33] proposed a simulation–optimization framework to simulate the behavior of counterfeiters and evaluate the effectiveness of various disruption strategies targeting counterfeit PPE supply chains during the COVID-19 pandemic. Magliocca et al. [42] presented an approach combining agent-based simulation models and spatial optimization techniques to study the dynamic interplay between counterdrug interdiction efforts and the adaptive responses of drug trafficking networks. The agent-based discrete event simulation model is used to simulate the highly dynamic, flexible, and ever-changing behavior of traffickers and their networks, while spatial optimization aims to maximize the coverage of estimated drug trafficking activities by deploying different types of enforcement resource packages at trafficking nodes. Based on a discrete event simulation model of a stylized counterfeit PPE supply chain, van Schilt et al. [62] investigated the effectiveness of various model calibration techniques, including Bayesian Computing, Bayesian Optimization, and Genetic Algorithms, in reconstructing the underlying structure of illicit supply chains under varying levels of data sparsity.

From the above review, it is evident that in the study of illicit trade and supply chain, very few studies have explored how to design efficient and secure parcel logistics systems to detect and reduce the influx of illicit goods or parcels into the country, especially focusing on the mail processing systems of postal services. Anzoom et al. [3] pointed out that, in addition to supply chain-level interdiction, effectively detecting physical illicit goods or parcels through the use of emerging inspection technologies and re-engineering inspection processes is also a significant research topic in illicit trade. Additionally, Anzoom et al. [3] suggested that the concepts and methods used in the multi-stage security screening systems for passenger transportation at airports [45,46,70,49,5,59,71] could be adapted and applied to the design of parcel logistics inspection systems and processes, with a focus on system security. Therefore, to the best of our knowledge, this study is the first to integrate the multi-stage security screening process used in airport passenger security systems into the design decisions of automated parcel logistics systems, aiming to efficiently and securely combat potentially illicit international parcels entering the country due to illegal trade.

2.2. Robotic sorting system

With the surge in online e-commerce transactions, the importance of labor cost savings achieved through automation in the warehouse picking process has grown, enabling parcel delivery companies and postal services to efficiently handle large volumes of parcels and provide convenient, low-cost next-day or same-day delivery services through advanced automated sorting systems [24,8]. In contrast to traditional conveyor-based sorting systems, the advanced robotic sorting systems developed in recent years feature a smaller footprint, lower costs, adaptable throughput control, and significantly greater flexibility and scalability, allowing for rapid adjustment of sorting operations by adding or removing robots based on parcel volume [8,21]. This has resulted in complex decision-making challenges at the tactical, strategic, and operational levels, which differ significantly from those in traditional automated sorting systems and require further investigation [8,21]. Despite the importance of these issues, the operations research community has largely overlooked robotic sorting systems, which represent the latest advancements in warehouse automation [8,10]. Boysen et al. [11] explored an order sorting system that includes tasks such as Piece-to-Order assignment and Order-to-Collection Point assignment. Their study, however, does not primarily focus on the scheduling and control of robots, and it assumes a constant travel time for robots to specific outlets. Shi et al. [53] investigated a human-robot hybrid sorting system that integrates both manual and robotic sorting processes, where the robotic sorting capacity is fixed, but manual labor can be scaled up or down based on demand. However, this study failed to fully leverage and explore the substantial inherent flexibility of the robotic system.

In addition to the two aforementioned articles considering variations

of the RSS system, the few remaining studies on robotic sorting systems focus on addressing the complex decision-making issues inherent to the original RSS system itself. These studies are particularly concentrated on strategic-level layout design/planning and operational-level decisions such as robot scheduling, robot routing, destination assignment and parcel assignment. Zi and Gao [72] and Zi and Gao [73] first investigated layout planning for parcel picking operations using autonomous robots. An M/M/s queueing model to estimate system throughput and formulated a mixed-integer nonlinear programming (MINLP) model are developed to optimize layout decisions such as aspect ratios, aisle widths, drop-off point and picking station placement, and their quantities for maximizing throughput and minimizing construction costs. Zou et al. [75] developed accurate closed queueing networks (CQN) to evaluate the performance of robot sorting systems, considering robot path topology and congestion. Using this closed queueing network model, this study analyzed the impact on throughput capacity and operational cost performance of different system layouts (including single-tier and two-tier layouts), various robot path topologies (such as shortest path and detour path topologies), and different robot-to-picking-station assignment rules (including random, closest, dedicated, and shortest-queue-based assignment rules) to identify indicate which RSS layout configuration is the better choice under specific conditions

Starostenko et al. [55] studied the impact of three different sorting space configuration models, including “E-type”, “Grid-type” (similar to “two-tier RSS with shortest path topology”) and “Football field” (similar to “Single-Tier RSS with shortest path topology”), on throughput performance. Zhao et al. [69] introduced a novel robot traffic management method, RC-S (Rhythmic Control for Sorting), tailored for RSS operations, and established an estimation formula integrating this robot traffic management method to describe the relationship between system performance and layout configuration. Subsequently, based on the performance analysis of RC-S, this study further presented a mixed integer linear programming (MILP) model for RSS layout design, considering RSS configuration, desired throughput, and costs, aiming to minimize expenses and determine the best system layout.

At the operational level, Zou and Chen [74] first investigated the impact of different parcel-to-picking-station assignment rules on throughput time using a constructed semi-open queueing network (SOQN). Tan et al. [56] developed a mixed-integer programming model and particle swarm optimization algorithm to determine the allocation of drop-off points to picking stations and the AMRs sorting schedule for minimizing total sorting time. Xu et al. [68] derived an open queueing network (OQN) incorporating conveyor congestion to estimate the RSS performance and then constructed an integer programming model to

Table 1

Summary of research differences/gaps between the existing RSS literature and this study.

Reference	System	Decision level	Decisions	AMR/Robot congestion	System security level	Layout type	Objective/Performance	Solution methods
Zou and Chen [74]	Robot parcel sortation	Operation planning	PAPS	✓	×	Grid-type	Throughput time	Semi-open queueing network
Zi and Gao [72]	Robot parcel sortation	Strategic planning	LFP	×	×	Grid-type	Throughput	M/M/s queueing model, Mixed integer nonlinear programming
Zi and Gao [73]	Robot parcel sortation	Strategic planning	LFP	×	×	Grid-type	Throughput	M/M/s queueing model
Shi et al. [53]	Human–Robot Hybrid Parcel sortation	Operational planning	NSP	✓	×	Grid-type	Total system cost	Analytic model
Tan et al. [56]	Robot parcel sortation	Operational planning	DAPS, AS	×	×	Grid-type	Makespan	Mixed integer programming, Particle swarm optimization
Zou et al. [75]	Robot parcel sortation	Strategic planning, Operational planning	LFP, AAPS	✓	×	Grid-type, Football field	Throughput, Total system cost	Closed queueing networks, Simulation
Xu et al. [68]	Robot parcel sortation	Tactical planning	DAPS	×	×	Grid-type	Throughput time	Open queueing network, Integer programming, Tabu search
Boysen et al. [11]	Robot order sortation	Operational planning	AS	×	×	—	Makespan	Mixed integer programming, Novel two-step multiple-scenario approach
Fang et al. [28]	Robot parcel sortation	Operational planning	AR, DADP	✓	×	Grid-type	Throughput	Semi-open queueing network, Mixed integer programming
Huang and Shen [35]	Robot parcel sortation	Operational planning	PAPS, AAPS, AR	×	×	Grid-type	Throughput	Multi-commodity min-cost network flow problem, Frank-Wolfe algorithm
Shen et al. [52]	Robot parcel sortation	Operational planning	DADP	×	×	Grid-type	Throughput	Multi-agent reinforcement learning
Starostenko et al. [55]	Robot parcel sortation	Strategic planning	LFP	×	×	Grid-type, Football field, E-type	Throughput	Simulation
Zhao et al. [69]	Robot parcel sortation	Strategic planning, Operational planning	LFP, AS	✓	×	Grid-type	Throughput, Total system cost	Autonomous vehicle management (Rhythmic Control), Mixed integer programming
Chen et al. [22]	Robot parcel sortation	Operational planning	DADP	✓	×	Grid-type	Throughput time	Open queueing network, Adaptive large neighborhood search
This study	Robot parcel sortation	Tactical planning	DAPS, NA, NIP	✓	✓	E-type	Total system cost	Mixed-integer stochastic optimization, Discrete Event Simulation, Mixed-integer simulation optimization algorithm (ORS-APHS)

Note. PAPS, parcel assignment to picking station; LFP, Layout and facility planning; NSP, number of sorting personnel; DAPS, drop-off point assignment to picking station; AS, AMR/Robot scheduling; AAPS, AMR/Robot assignment to picking station; AR, AMR/Robot routing; DADP, destination assignment to drop-off point; NA, number of AMRs/Robots; NIP, number of inspection personnel; IPD, inspection probability determination.

optimize the assignment of parcels to picking stations for minimizing throughput time. Fang et al. [28] employed a semi-open queuing network model for the RSS system, considering robot movement dynamics and network congestion, to study how dynamic operational decisions integrating AMR/robot routing and destination assignment to drop-off points affect system performance. Next, this study utilized the structural properties of Markov decision processes to derive the optimal dynamic robot routing policy and constructed a MILP model to determine the optimal destination reassignment schemes aimed at minimizing system workload imbalance. Huang and Shen [35] formulated integrated robot-to-picking-station allocation, parcel-to-picking-station assignment, and pathfinding problems as a multi-commodity minimum-cost network flow problem and applied the Frank-Wolfe algorithm to derive the approximated system-optimal traffic assignment and AMR path planning solution, demonstrating superior performance over fast multi-agent path planning algorithms. Although strategic and operational-level decision-making issues in RSS systems have begun to be explored, to the best of our knowledge, tactical-level resource allocation decisions, such as the number of AMRs, the number of inspection personnel, and the allocation of drop-off points at each picking station, as well as system security settings to counter illicit trade in international parcel logistics, have never been investigated. Table 1 summarizes the differences between the existing RSS literature and this study.

2.3. Mixed-Integer simulation optimization

As shown in Table 1, the queueing models such as SOQN [74,28], M/M/S [72,73], CQN [75], and OQN [68] are common modeling tools to consider the stochastic and dynamic behavior of RSS systems to conduct performance evaluations and optimization decisions. To obtain analytical solutions, these queueing models always make several impractical assumptions without considering any kind of stochastic processes and complex interdependent dynamic properties in actual RSS systems, such as the probability distribution of parcel arrivals, drop-off point distributions, AMR acceleration and deceleration, parcels waiting on conveyor belts for picking stations, AMRs queuing for loading at picking stations, and AMR/robot transport congestion, etc. Meanwhile, simulation optimization has emerged as a powerful and flexible tool over the past few years, finding success in numerous practical and complex applications [57,30,2,17,29]. By creating high-resolution stochastic simulation models that capture the stochastic, dynamic and complex behaviors of real-world systems, simulation optimization algorithms can generate more suitable and practical optimal decisions compared to those derived from queueing models [54,76].

According to relevant simulation optimization review literature [57,30,2,17,29], previous studies have proposed various specifically designed algorithms, but most have focused on addressing simulation optimization problems with single-variable types, such as continuous [57,13,14,16] or integer decision variables [34,67,15]. These algorithms are not suitable for the tactical design problem of robotic sorting systems with guaranteed security levels, as this problem involves a mix of continuous and integer decision variables. The mixed nature of the solution space and the computationally expensive simulation processes make such problems highly challenging. In recent years, only a few studies have focused on mixed-integer simulation optimization problems [63,41,64,60]. However, these approaches often assume a “smoothness” structure, making them unsuitable for problems lacking performance function gradients, such as the tactical design decisions of the RSS system [63,41,64]. Furthermore, these algorithms fail to consider effective allocation of simulation replications to accelerate the search process, resulting in wasted simulation resources on evaluating poor solutions [60]. Although metaheuristic algorithms, such as Genetic Algorithms (GA) and Tabu Search (TS), are widely applied to large-scale simulation optimization problems, they exhibit notable drawbacks [2,26,17]. These include being applicable only to single-variable types (e.g., GA is better suited for discrete variable problems), lacking

problem-specific global and local search mechanisms, and similarly failing to address the effective allocation of simulation replications, leading to inefficient use of simulation resources [26]. To address the limitations of existing research, this study proposes the ORS-APHS algorithm, which leverages an innovative initialization method, a neighboring solution structure that integrates global exploration and local exploitation, and an efficient simulation resource allocation mechanism. This ORS-APHS algorithm not only efficiently solves the tactical design problem of RSS system with a guaranteed security level but also can be extended to other stochastic optimization problems involving mixed continuous and integer variables.

2.4. Key contributions

Based on the aforementioned analysis, the main contributions of this paper are outlined below.

1. We investigate the tactical design decisions for a robotic sorting system with guaranteed security level, which incorporates multi-stage security screening processes to effectively and securely intercept potentially illicit international parcels resulting from illegal trade. Furthermore, a mixed-integer stochastic optimization model is constructed to derive tactical design optimal solutions, including the number of required AMRs, the number of inspection personnel, the allocation of drop-off points at each picking station, and system security settings, to minimize overall expected system costs.
2. Since the objective function of the mixed-integer stochastic optimization model is not analytically available, a high-resolution discrete event simulation model considering the detailed stochastic and dynamic nature of the RSS system with a guaranteed security level is first constructed to estimate this performance measure. Then, we develop an efficient mixed-integer simulation optimization algorithm, namely ORS-APHS. This algorithm integrates OCBA-based ranking and selection (integer variables for the number of required AMRs, the number of inspection personnel and the drop-off point allocation) with adaptive particle and hyperball search (continuous variables for system security settings) to effectively find the optimal tactical design solution for the RSS system with a guaranteed security level over mixed continuous and integer feasible domains under limited simulation budgets.
3. We collaborated with Chunghwa Post in Taiwan to conduct extensive numerical experiments using the design of the international import parcel sorting system in their new mail processing center as a case study, demonstrating the efficiency and effectiveness of the proposed ORS-APHS algorithm. Through sensitivity analysis, we investigated the impact of changes in system security levels, variations in the number of available AMRs, adjustments in physical re-inspection times, and distributions of different drop-off points for arriving parcels on total system costs, various cost types, and tactical design decisions. The insights gained from these results will equip the decision makers with the tools to make informed tactical design and resource allocation decisions, ultimately leading to the development of an efficient and secure RSS system.

3. Tactical design decisions of robotic sorting system with guaranteed security level

3.1. Problem description and definition

A robotic sorting system with a guaranteed security level is primarily used for handling international import operations by postal service systems or international parcel logistics companies [22,70]. When inbound trucks deliver parcels from international imports to the parcel sorting logistics center, the robotic sorting system first conducts a legality inspection of the parcel contents. After the inspection, each parcel is sorted to different drop-off points according to its destination, with

each drop-off point representing a specific destination. Once all parcels have been sorted, each drop-off point will send the collected parcels to its corresponding outbound trucks for loading, which then transport the parcels to their respective destinations according to the outbound scheduling timetable.

Apart from the inbound and outbound truck operations, the main components of the robot sorting system with guaranteed security level include international parcels with destinations, a cross-belt conveyor with an induction station, multi-stage security inspection operations, multiple drop-off points, several picking stations, and a fleet of AMRs [22,70], as shown in Fig. 1. Each international parcel has a fixed destination, and the purpose of the sorting operation is to consolidate parcels with the same destination at a specific drop-off point for subsequent outbound distribution. The cross-belt conveyor is used to transport parcels through the inspection station and then divert them to different picking stations for sorting and transportation by AMR. The induction station is used to feed parcels onto the cross-belt conveyor through robotic arms and identify the destination information of the parcels, which is used for sorting. The multi-stage security inspection operation consists of the initial screening station at first stage (known as stage 1) and the thorough physical re-inspection station at second stage (known as stage 2). The initial screening station mainly uses X-ray machines to conduct preliminary inspections of parcels. If suspicious illegal parcels are found, they will be sent to the physical re-inspection station for physical parcel unpacking and re-inspection by customs officers. If the parcels are indeed illegal after re-inspection, they will be destroyed; otherwise, they will be transported to the subsequent AMR sorting operation through the cross-belt conveyor. The purpose of the

drop-off point is to collect parcels with the same destination together. After the sorting operation is completed, these parcels will be bundled into bags or loaded into carts and sent to the corresponding outbound truck. This study assumes that each drop-off point is only assigned to one specific destination, therefore, in this study, drop-off points and parcel destinations are synonymous. The picking station is equipped with a chute for receiving parcels from the cross-belt conveyor and a robotic arm or picker for placing parcels onto the AMR. The AMR fleet is used to handle parcel transportation from the picking station to the drop-off point. The AMR control in this study adopts a dedicated strategy [75], meaning that each AMR serves specific picking stations. After the AMR completes the sorting and transportation, it will return to the waiting area in front of its serviced picking stations to wait for the next sorting and transportation task.

Based on the system configuration mentioned above, and given the “E-type” layout configuration [55] of the RSS system (including robot path directions, size of the sorting area, location and number of picking stations, location of drop-off points etc.), the tactical design decisions of the robotic sorting system with guaranteed security level aim to determine the number of required AMRs at each picking station, the number of re-inspection personnel, the allocation of drop-off points at each picking station, and the security screening probabilities at each inspection stage. The main goal is to minimize the expected overall system cost, including the total acquisition cost of AMRs, the total personnel cost for the physical re-inspection station, and the expected system operational cost of RSS, while satisfying specified security threshold constraints. However, the biggest challenge in addressing the tactical design problem is its mid-term nature (e.g., monthly or quarterly

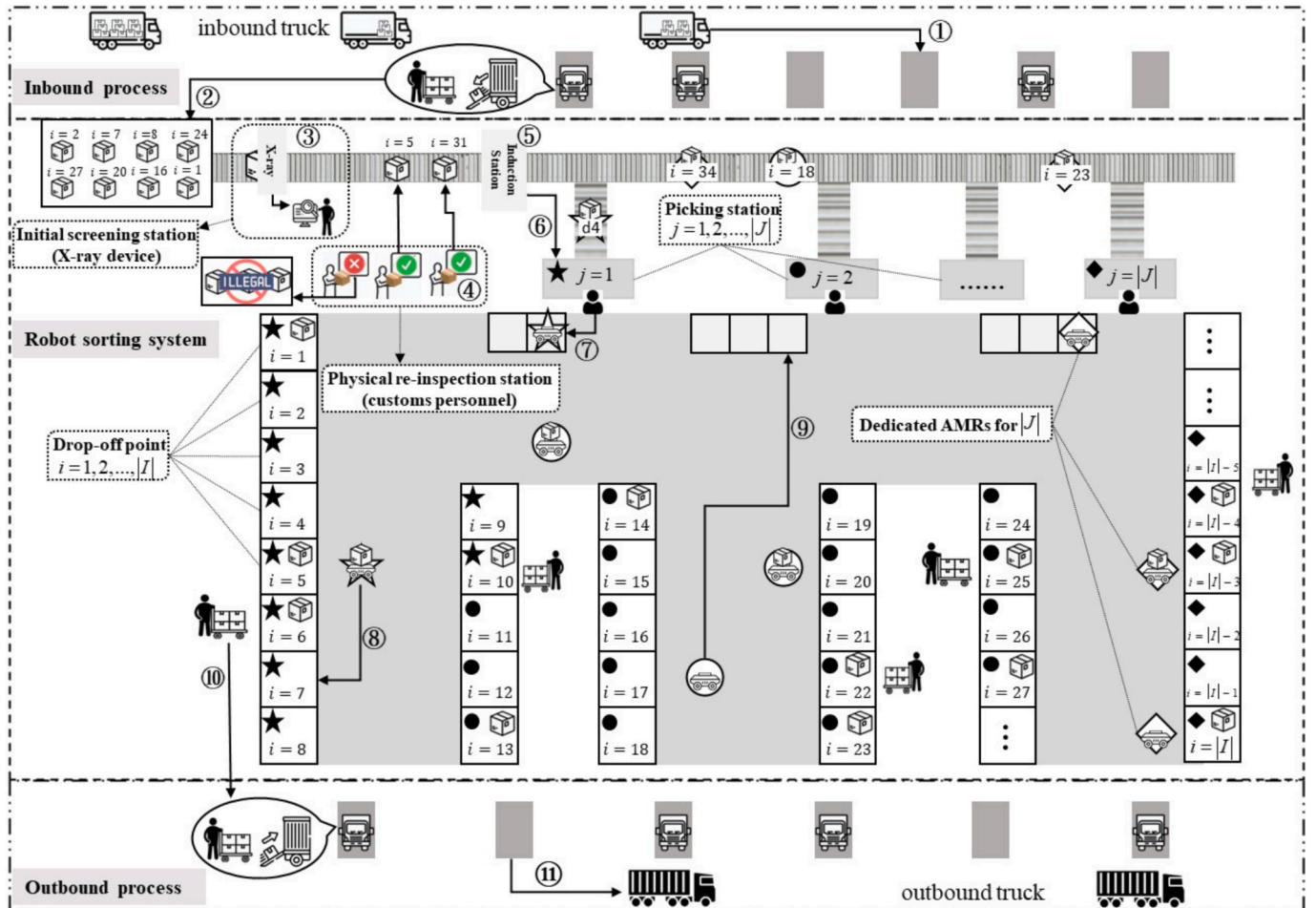


Fig. 1. Composition elements, layout, and overall operational process of a robotic sorting system with guaranteed security level.

adjustments), which necessitates accounting for the dynamic and stochastic characteristics of the RSS system (including the arrival quantity of parcels, time-varying stochastic parcel arrivals, mixing ratio of parcels for different destinations, stochastic re-inspection time, AMR waiting time, AMR travel time, etc.) to make robust decisions [68,55,69]. Therefore, this study develops a stochastic optimization model and an efficient mixed-integer simulation optimization algorithm to solve the problem.

Prior to modeling the tactical design problem for the robotic sorting system with a guaranteed security level, several assumptions need to be clarified as follows.

1. The strategic layout configuration of the RSS system with guaranteed security level, including the robot path directions, the size of the sorting area, the number and location of picking stations, the size of the AMR waiting area at each picking station, and the location of drop-off points, is known and given.
2. Each drop-off point is assigned to a specific destination, and different drop-off points represent different destinations. As a result, in this study, drop-off points and parcel destinations are synonymous.
3. AMRs operate under a dedicated control strategy [75], where each AMR is assigned to particular picking stations. After completing its sorting and transportation duties, the AMR returns to the waiting area adjacent to its designated picking stations, ready for the next task.
4. A picking station can process parcels for multiple destinations, but parcels for the same destination (drop-off point) can only be processed by a specific picking station.
5. To account for the dynamic and stochastic nature of the RSS system as well as potential AMR congestion, we have developed a high-resolution stochastic simulation model and an efficient ORS-APHS algorithm to search and determine the best tactical design decisions. In building this simulation model, we assume that AMRs utilize the shortest path algorithm for their operational-level route planning.

3.2. Mixed-integer stochastic optimization model

In this section, the tactical design problem for a robotic sorting system with guaranteed security level is formulated as a mixed-integer stochastic optimization model. The indices, input parameters, and decision variables of this model are defined as follows.

Sets and Indices

- I : Set of parcel destinations/ drop-off point, indexed by i
 J : Set of parcel picking stations, indexed by j

Parameters

- s_j : Maximum number of drop-off points served by picking station j
 a_j : Maximum number of AMRs allowed to be parked at picking station j
 d_{ij} : Shortest delivery distance from drop-off point i to picking station j
 l_i : The expected number of parcels to be sorted to drop-off point i
 c_1 : Setup or acquisition cost of one AMR
 c_2 : Personnel cost in the thorough physical re-inspection station at stage 2
 c_3 : Operational cost per unit of time for the RSS system
 r : Maximum number of AMRs that can be used in the RSS system
 $\bar{n}$: Maximum number of personnel at the thorough physical re-inspection station at stage 2
 $\underline{n}$: Minimum number of personnel at the thorough physical re-inspection station at stage 2
 θ : Prespecified threshold of the system security level
 A : An event representing parcels entering the AMRs sorting operation after the inspection is completed
 T : An event representing illegally threatening parcels

Decision Variables

- P_1 : Proportion of parcels selected for thorough physical re-inspection at stage 2 after the initial screening at stage 1

(continued on next column)

(continued)

P_2 :	Proportion of illegal parcels intercepted by the re-inspection personnel at the thorough physical re-inspection station. This indicates the thoroughness of the parcel inspection conducted by the re-inspection personnel.
N :	Number of re-inspection personnel at the thorough physical re-inspection station
X_j :	Number of AMRs assigned to picking station j
$\mathbf{X}$:	Matrix representing number of AMRs assigned to each picking station $\mathbf{X} = (X_j)_{ J }$
Y_{ij} :	Determine whether drop-off point i is assigned to picking station j for sorting
$\mathbf{Y}$:	Matrix representing the allocation of all drop-off points to each picking station $\mathbf{Y} = (Y_{ij})_{ I \times J }$

Objective Function

$$\text{Minimize}_{P_1, P_2, N, \mathbf{X}, \mathbf{Y}} c_1 \times \sum_{j=1}^J X_j + c_2 \times N + c_3 \times \mathbf{E}[OT(P_1, P_2, N, \mathbf{X}, \mathbf{Y}; \omega)] \quad (1)$$

Constraints

$$\sum_{j=1}^J Y_{ij} = 1 \quad \forall i \in I \quad (2)$$

$$\sum_{i=1}^I Y_{ij} \geq 1 \quad \forall j \in J \quad (3)$$

$$\sum_{i=1}^I Y_{ij} \leq s_j \quad \forall j \in J \quad (4)$$

$$1 \leq X_j \leq a_j \quad \forall j \in J \quad (5)$$

$$\sum_{j=1}^J X_j \leq r \quad (6)$$

$$\underline{n} \leq N \leq \bar{n} \quad (7)$$

$$\mathbf{P}\{T|A\} \leq \theta \quad (8)$$

$$0 \leq P_1, P_2 \leq 1 \quad (9)$$

$$N \in \mathbf{Z}^+ \quad (10)$$

$$X_j \in \mathbf{Z}^+ \quad \forall j \in J \quad (11)$$

$$Y_{ij} \in \{0, 1\} \quad \forall i \in I, j \in J \quad (12)$$

Eq. (1) is the objective function which minimizes total cost in the mobile robotic sorting system, which includes total acquisition cost of AMRs, total personnel cost for the physical re-inspection station at stage 2, and the expected system operational cost of RSS. The expected system operational cost is calculated based on the expected operational time, $\mathbf{E}[OT(P_1, P_2, N, \mathbf{X}, \mathbf{Y}; \omega)]$, required by the RSS to complete all parcel sorting tasks. The operational time, $OT(P_1, P_2, N, \mathbf{X}, \mathbf{Y}; \omega)$, is a stochastic function which depends on the decision variables, $P_1, P_2, N, \mathbf{X}, \mathbf{Y}$, of tactical design decisions in RSS and randomness effect ω caused by stochastic arrival quantity of parcels, time-varying stochastic parcel arrivals, mixing ratio of parcels for different destinations, stochastic re-inspection time at stage 2, stochastic AMR travel time. Because the expected operational time, $\mathbf{E}[OT(P_1, P_2, N, \mathbf{X}, \mathbf{Y}; \omega)]$, is not analytically available, a stochastic RSS simulation model with multiple replications is developed to estimate this performance measure. Constraint (2) ensures that each drop-off point is assigned to only one picking station for the sorting service. Constraint (3) shows that each picking station is responsible for at least one drop-off point for sorting service. Constraint (4) indicates that the number of drop-off points assigned to each picking station cannot exceed the maximum service limit. Constraint (5) states that each picking station must be assigned at least one AMR for the

sorting operation, but the total number of AMRs assigned cannot exceed the maximum parking capacity allowed for this picking station. Constraint (6) ensures that the total number of AMRs assigned to all picking stations does not exceed the maximum limit of AMRs allowed for use in the RSS system. Constraint (7) specifies the upper and lower bounds for the number of re-inspectors in the physical re-inspection station. Constraint (8) guarantees that the system security level does not exceed the pre-specified threshold θ . This means that the probability

$$\mathbf{P}\{T|A\} = \frac{\mathbf{P}\{A \cap T\}}{\mathbf{P}\{A \cap T\} + \mathbf{P}\{A \cap T^c\}} \quad (13)$$

From the above equation, it is known that calculating the system security level only requires finding $\mathbf{P}\{A \cap T\}$ and $\mathbf{P}\{A \cap T^c\}$. As shown in Fig. 2, We define the following events to classify the parcels who enter service:

- $S_{1d} = \{\text{Set of parcels selected for further physical re-inspection through initial screening at stage1}\}$ (S_1 is the complement of S_{1d})
- $S_{1d}^{T^c} = \{\text{Set of legal parcels sent to the physical re-inspection station at stage2}\}$
- $S_{1d}^T = \{\text{Set of illegal parcels sent to the physical re-inspection station at stage2}\}$
- ($S_{1d}^{T^c}$ is the complement of S_{1d}^T)
- $S_{2d} = \{\text{Set of illegal parcels detected at the physical re-inspection station}\}$
- $S_{2d}' = \{\text{Set of undetected illegal parcels entering AMRs sorting operation}\}$
- (S_{2d}' is the complement of S_{2d})
- $A = S_1 \cup S_{1d}^{T^c} \cup S_{2d}' = \{\text{Set of all parcels entering AMRs sorting operation}\}$

of illegal parcels appearing among all parcels processed by the RSS system for picking operations, after passing through all inspection procedures, does not exceed the pre-specified threshold θ , $\mathbf{P}\{T|A\} \leq \theta$. In Section 3.3, we will explain how to calculate the system security level and its property. Constraints (9)-(12) define the range and type of decision variables, including continuous variables P_1, P_2 , integer variables $N, \mathbf{X}$ and binary variables $\mathbf{Y}$. To efficiently tackle this complicated stochastic optimization model with mixed continuous and integer variables, we propose a novel mixed-integer simulation optimization algorithm, called ORS-APHS, to search for the optimal solutions of multiple decision variables, $P_1, P_2, N, \mathbf{X}, \mathbf{Y}$, over mixed continuous, integer and binary feasible domains under limited simulation budgets. More details regarding the ORS-APHS algorithm design are presented in Section 4.

3.3. System security level

In this section, we first compute the system security level $\mathbf{P}\{T|A\}$ and then demonstrate its monotonicity. Through Bayes' theorem, the system security level $\mathbf{P}\{T|A\}$ can be written as follows:

Considering the probability of illegal parcels being a threatening risk is $\mathbf{P}\{T\} = \tau$ and define the probability that the illegal parcels selected to the physical re-inspection station through initial screening at stage 1 indeed pose a threat as $\mathbf{P}\{T|S_{1d}\} = \beta$. For an effective initial screening conducted in stage 1, it is reasonable to assume that $\beta > \tau$. First, by the law of total probability, the threatening probability of illegal parcels τ can be expressed as follows:

$$\begin{aligned} \tau &= \mathbf{P}\{T\} = \mathbf{P}\{T|S_1\} \times \mathbf{P}\{S_1\} + \mathbf{P}\{T|S_{1d}\} \times \mathbf{P}\{S_{1d}\} + \mathbf{P}\{T|S_{2d}'\} \times \mathbf{P}\{S_{2d}'\} + \mathbf{P}\{T|S_{2d}\} \times \mathbf{P}\{S_{2d}\} \\ &= \mathbf{P}\{T|S_1\} \times (1 - P_1) + \mathbf{P}\{T|S_{1d}\} \times P_1 \times \beta \times (1 - P_2) + \mathbf{P}\{T|S_{2d}'\} \times P_1 \times \beta \times P_2 + \mathbf{P}\{T|S_{2d}\} \times P_1 \times \beta \times P_2 \end{aligned} \quad (14)$$

Because $\mathbf{P}\{T|S_{2d}'\} = \mathbf{P}\{T|S_{2d}\} = 1$, we can derive the following equation:

$$\mathbf{P}\{T|S_1\} = \frac{\tau - P_1 \times \beta}{1 - P_1} \quad (15)$$

With the above preparations and applying the law of total probability again, we can respectively calculate $\mathbf{P}\{A \cap T\}$ and $\mathbf{P}\{A \cap T^c\}$ as follows.

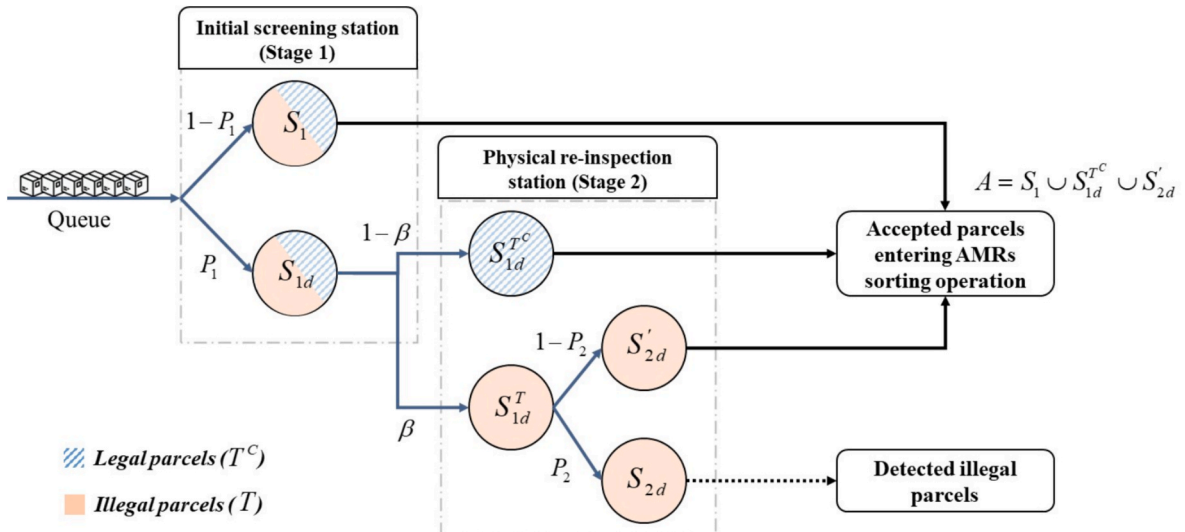


Fig. 2. An illustration of the status and proportion of parcels selected for initial screening and physical re-inspection stations.

$$\begin{aligned}
\mathbf{P}\{A \cap T\} &= \mathbf{P}\{A \cap T \cap S_1\} + \mathbf{P}\{A \cap T \cap S_{2d}^c\} \\
&= \mathbf{P}\{A|T \cap S_1\} \times \mathbf{P}\{T|S_1\} \times \mathbf{P}\{S_1\} + \mathbf{P}\{A|T \cap S_{2d}^c\} \times \mathbf{P}\{T|S_{2d}^c\} \times \mathbf{P}\{S_{2d}^c\} \\
&= \tau - P_1 \times \beta \times P_2
\end{aligned} \tag{16}$$

$$\begin{aligned}
\mathbf{P}\{A \cap T^c\} &= \mathbf{P}\{A \cap T^c \cap S_{1d}^{Tc}\} + \mathbf{P}\{A \cap T^c \cap S_1\} \\
&= \mathbf{P}\{A|T^c \cap S_{1d}^{Tc}\} \times \mathbf{P}\{T^c|S_{1d}^{Tc}\} \times \mathbf{P}\{S_{1d}^{Tc}\} + \mathbf{P}\{A|T^c \cap S_1\} \times \mathbf{P}\{T^c|S_1\} \times \mathbf{P}\{S_1\} \\
&= 1 - \tau
\end{aligned} \tag{17}$$

Based on the above Eqs. (16) and (17), we can derive Eq. (18) to calculate the system security level $\mathbf{P}\{T|A\}$ of the RSS system. Moreover, this system security level is a monotonically decreasing function with respect to P_1 and P_2 .

$$\mathbf{P}\{T|A\} = \frac{\mathbf{P}\{A \cap T\}}{\mathbf{P}\{A \cap T\} + \mathbf{P}\{A \cap T^c\}} = \frac{\tau - P_1 \times \beta \times P_2}{1 - P_1 \times \beta \times P_2} \tag{18}$$

Proposition 1. ($\mathbf{P}\{T|A\}$ is a decreasing function with respect to P_1 and P_2 .) *Proof.* See Section A1 of Appendix A.

Intuitively, it is commonly believed that the expected goal of the mobile robotic sorting system in the illicit supply chain should be to minimize $\mathbf{P}\{T|A\}$ to enhance the security of the RSS system. Through the results of Proposition 1, minimizing $\mathbf{P}\{T|A\}$ can be achieved by choosing sufficiently high P_1 and P_2 , because P_1 and P_2 are controllable in the tactical design decisions of the RSS system. However, higher values of P_1 and P_2 will lead to increased operation time and more inspection personnel needed at the physical re-inspection station, thereby raising the expected system operating costs. Therefore, there is a trade-off relationship between the system security level and the system operating costs. After discussions with practitioners, it was concluded that the security level must be maintained above a specific threshold before considering the goal of minimizing system operating costs. This involves converting the system security level $\mathbf{P}\{T|A\}$ into a constraint, similar to the concept of a security-check system [51,71]. Consequently, determining the appropriate values of P_1 and P_2 to ensure that the tactical design of the RSS system minimizes total cost while maintaining a certain level θ of system security is the main objective of this study. Based on the Eqs. (8) and (18), the following constraint (19) is derived to determine the feasible solution range for P_1 and P_2 , and serve as the basis for the simulation optimization algorithm design in Section 4.

$$P_1 \times P_2 \geq \frac{\tau - \theta}{\beta \times (1 - \theta)} \tag{19}$$

4. ORS-APHS algorithm

To solve this complex stochastic optimization model (1)-(13) with an expected objective function and mixed continuous, integer, and binary decision variables, we first significantly reduce the number of decision variables by replacing the original binary variables Y_{ij} with new integer variables Z_i . Since each drop-off point is only served by one picking station (see Constraint (2)), Z_i , where $Z_i \in J = \{1, 2, \dots, |J|\}$, represents which picking station drop-off point i is assigned to for sorting and Z is a matrix representing the allocation of all drop-off points. For example, $Z_{100} = 3$ indicates that the 100th drop-off point is assigned to the 3rd picking station.

Then, for the RSS tactical design problem with mixed continuous and integer decision variables obtained through the above transformation, we propose a mixed-integer simulation optimization approach called ORS-APHS. This method integrates OCBA-based ranking and selection (for integer variables N, X, Z) with adaptive particle and hyperball search (for continuous variables P_1, P_2) to effectively find the optimal tactical design solution for the RSS system with a guaranteed security level over

mixed continuous and integer feasible domains. In ORS-APHS, the candidate solution indexed as b is composed of the integer solutions $S_b = (N, X, Z)_b$ and the continuous solutions $P_b = (P_1, P_2)_b$. For a given solution (S_b, P_b) , $\overline{TC}(S_b, P_b)$ represents the estimated objective value to approximate objective function (1) according to multiple simulation replications of the RSS discrete event simulation model. The main characteristics of ORS-APHS are described below. ORS-APHS consists of several "APHS + OCBA- m based Screening" iterations $r = 1, 2, \dots, R - 1$ and one final selection iteration R . In the initial iteration $r = 1$, we develop the greedy heuristic method to generate B promising initial integer solutions $\{S_1, S_2, \dots, S_B\}$ and randomly specify their corresponding continuous initial solutions $\{P_1, P_2, \dots, P_B\}$ satisfying the constraint (19). Each solution is assigned an index based on the order in which it is created during the initialization process. This set of indexes of these initial solutions is stored in $M(1)$. During other r th iteration ($r > 1$), B many candidate integer solutions are produced by the heuristic neighborhood rules from the neighborhood of promising solutions and the indexes of these newly generated solutions are stored to $M(r)$. Each candidate solution is indexed based on the order in which it appears during the neighborhood search process. The index set of surviving solutions after iteration $r - 1$ is denoted $V(r - 1)$. For each iteration r , we first performs the APHS algorithm with global and local search mechanisms to obtain the optimal continuous solution P_b^* corresponding to each integer solution $\{S_b, b \in \{M(r) \cup V(r - 1)\}\}$ over the feasible region defined by constraint (19). Then, an objective screening procedure combining the OCBA- m algorithm with the RSS simulation model is adopted to eliminate inferior solutions and filter out the top m best surviving solutions. The indexes of the surviving solutions resulting from this OCBA- m based objective screening are stored in $V(r)$ after iteration r . After $R - 1$ screening iterations, the final selection stage identifies the best solution (S^*, P^*) from final surviving solutions $\{(S_b, P_b^*), b \in V(R - 1)\}$ in terms of the objective measure and obtains the optimal objective value $\overline{TC}(S^*, P^*)$. A high-level description of ORS-APHS is presented below, with the detailed pseudo code shown in Table 2.

ORS-APHS algorithm

Setup: Specify maximum number of iterations R , maximum number of solutions B of each iteration, number of survival solutions m of each iteration, replenishment probability RP , total simulation budgets TS , desired PCS $P\{CS\}_{desired}$, and initial simulation replications n_0 . Set the iteration number $r = 1$, the initial survival set $V(0) = \phi$ and the initial cumulative simulation replications $cs = 0$.

Initialization: Generate B initial integer solutions $\{S_1, S_2, \dots, S_B\}$ using a greedy heuristic method (see Section 4.1), and randomly specify their corresponding continuous initial solutions $\{P_1, P_2, \dots, P_B\}$ satisfying the constraint (19), where $S_b = (N, X, Z)_b$, $P_b = (P_1, P_2)_b$ and $b \in M(1) = \{1, 2, \dots, B\}$. Go to Adaptive Particle and Hyperball Search Stage.

Adaptive Particle and Hyperball Search: For all S_b , where $b \in \{M(r) \cup V(r - 1)\}$, apply APHS algorithm (see Section 4.2) and discrete event simulation model of the RSS system to find their corresponding optimal continuous solutions P_b^* over the feasible region defined by constraint (19) and return the number of simulation replications used by the APHS algorithm to update cs . Go to OCBA- m based Objective Screening Stage.

OCBA- m based Objective Screening Stage: For all (S_b, P_b^*) , where $b \in \{M(r) \cup V(r - 1)\}$, use OCBA- m algorithm (see Section 4.3) and discrete event simulation model of the RSS system to filter out the top m surviving solutions, put their corresponding index into the set of surviving solutions $V(r)$, and return the number of simulation replications used by the OCBA- m algorithm to update cs . Go to Searching and Stopping Stage.

Searching and Stopping Stage: Update $r = r + 1$. If $r < R$ or $cs < TS$, then generate B many new integer solutions from the neighborhood of the surviving solutions $\{S_b, b \in V(r - 1)\}$ with

Table 2

Pseudo code of ORS-APHS algorithm.

Input: $R, B, m, RP, TS, P\{CS\}_{desired}, n_0$.	
Output: The best solution (S^*, P^*) and the optimal objective value $\overline{TC}(S^*, P^*)$.	
1	Setup: Set the iteration number $r = 1$, the initial survival set $V(0) = \emptyset$ and the initial cumulative simulation replications $cs = 0$.
Initialization: Generate B initial integer solutions $\{S_1, S_2, \dots, S_B\}$ using a greedy heuristic method (see Section 4.1) and randomly	
2	specify their corresponding continuous initial solutions $\{P_1, P_2, \dots, P_B\}$. Let $S_b = (N, X, Z)_b$, $P_b = (P_1, P_2)_b$ and $b \in M(1) = \{1, 2, \dots, B\}$
3	For $b = 1$ to $ M(r) \cap V(r-1) $ do
4	Select and fix the solution S_b .
5	Apply APHS algorithm (see Section 4.2) and discrete event simulation model of the RSS system to find their corresponding optimal
6	continuous solutions P_b^* over the feasible region defined by constraint (19).
7	Update cs using the number of simulation replications used by the APHS algorithm.
8	Let $b = b + 1$.
9	End
10	Filter out the top m surviving solutions from all (S_b, P_b^*) , where $b \in \{M(r) \cup V(r-1)\}$ and put their corresponding index into
11	the set of surviving solutions $V(r)$ by OCBA- m algorithm (see Section 4.3).
12	Update cs using the number of simulation replications used by the OCBA- m algorithm.
13	Let $r = r + 1$.
14	While $r < R$ or $cs < TS$ do
15	For $b = 1$ to B do
16	Generate a random number $\rho \sim Uniform(0, 1)$.
17	If $\rho < RP$ then
18	Generate a new candidate solution from the neighborhood of the surviving solutions $\{S_b, b \in V(r-1)\}$ using the
19	heuristic neighborhood rules (see Section 4.4).
20	Else
21	Generate a new candidate solution using the initialization strategy (the greedy heuristic method).
22	End
23	Update $M(r)$ by adding new candidate solution.
24	Let $b = b + 1$.
25	End
26	For any S_b , where $b \in \{M(r)\}$, that has been visited during previous iterations, retrieve its corresponding latest P_b^* from the
27	previous APHS iterations.
28	For the S_b that has not been visited, randomly specify its corresponding continuous solution P_b satisfying the constraint (19).
29	For $b = 1$ to $ M(r) \cap V(r-1) $ do
30	Select and fix the solution S_b .
31	Apply APHS algorithm (see Section 4.2) and discrete event simulation model of the RSS system to find their corresponding
32	optimal continuous solutions P_b^* over the feasible region defined by constraint (19).
33	Update cs using the number of simulation replications used by the APHS algorithm.
34	Let $b = b + 1$.
35	End
36	Filter out the top m surviving solutions from all (S_b, P_b^*) , where $b \in \{M(r) \cup V(r-1)\}$ and put their corresponding index
37	into the set of surviving solutions $V(r)$ by OCBA- m algorithm.
38	Update cs using the number of simulation replications used by the OCBA- m algorithm.
39	Let $r = r + 1$.
40	End
41	Identify and return the best solution (S^*, P^*) within the surviving solutions $\{(S_b, P_b^*), b \in V(R-1)\}$ and obtain the optimal
42	objective value $\overline{TC}(S^*, P^*)$.

replenishment probability RP using the heuristic neighborhood rules (see Section 4.4), update $M(r)$, and go to **Counter Tracking Stage**. If $r = R$ or $cs \geq TS$, go to **Selection Stage**.

Counter Tracking Stage: For any S_b , where $b \in \{M(r)\}$, that has been visited during previous iterations, retrieve its corresponding latest P_b^* from the previous APHS iterations. For the S_b that has not been visited, randomly specify its corresponding continuous initial solution P_b satisfying the constraint (19). Go back to **Adaptive Particle and Hyperball Search Stage**.

Selection Stage: Identify the best solution (S^*, P^*) within the

surviving solutions $\{(S_b, P_b^*), b \in V(R-1)\}$ and obtain the optimal objective value $\overline{TC}(S^*, P^*)$.

4.1. Generation of the initial solutions

The quality of the initial solutions is relatively important. Good initial solutions can enable the algorithm to quickly focus on promising areas. Deep exploitation in the neighborhood of promising surviving solutions is also likely to produce high-quality candidate solutions. Consequently, based on the characteristics of the tactical design problem

of the RSS system, this study develops a greedy heuristic method for integer decisions S_b to generate higher-quality initial solutions, including drop-off point assignment solution Z , AMRs allocation solution for picking stations X , and re-inspection personnel allocation solution N . First, the initialization strategy for drop-off point assignment is based on the principle that drop-off points are more likely to be assigned to closer picking stations. For example, if the distance between drop-off point 1 and picking station 1 is 5 m, while the distance between drop-off point 1 and picking station 2 is 10 m, then picking station 1 is more likely to be assigned to drop-off point 1 than picking station 2. Then, based on the results of this drop-off point assignment and the expected parcel demand for each drop-off point, the initialization strategy for AMR allocation decision for picking stations is to allocate the required number of AMRs proportionally according to the total parcel demand assigned to each picking station. For example, if there are 10 AMRs to be allocated, and picking station 1 has to sort 100 parcels in total while picking station 2 has to sort 400 parcels, then picking station 1 would receive 2 AMRs and picking station 2 would receive 8 AMRs. The detailed procedure for generating the initial solution is described as follows:

Step 1. For all drop-off points, calculate the shortest delivery distance d_{ij} from each drop-off point i to each picking station j by the shortest path algorithm. We can obtain traveling distance matrix $\{d_{11}, d_{12}, \dots, d_{1|J|}, d_{21}, d_{22}, \dots, d_{2|J|}, \dots, d_{ij}, \dots, d_{i|J|}, d_{i+1,1}, d_{i+1,2}, \dots, d_{i+1|J|}, \dots, d_{|I|1}, d_{|I|2}, \dots, d_{|I||J|}\}$ from all drop-off points to all picking stations. Let the set PS_j to store which drop-off point is allocated in picking station j . Set $i = 1$.

Step 2. Calculate total distance $TD = \sum_j d_{ij}$ of this drop-off point i .

Step 3. Calculate the difference values td_{ij} between the total distance and the distance from drop-off point i to picking station j using $td_{ij} = TD - d_{ij}$ and the sum of total difference value $NTD = \sum_j td_{ij}$.

Step 4. Calculate the probability value pc_{ij} of each picking station j being selected using $pc_{ij} = \frac{td_{ij}}{NTD}$. We can obtain the probability values $\{pc_{i1}, pc_{i2}, \dots, pc_{i|J|}\}$ of this drop-off point selecting all picking stations.

Step 5. Calculate the cumulative probability interval of each picking station j . If $j = 1$, the cumulative probability interval is $(0, pc_{i1}]$. Otherwise, the cumulative probability interval of each picking station j ($j > 1$) is $(\sum_{j=1}^{j-1} pc_{ij}, \sum_{j=1}^j pc_{ij}]$.

Step 6. Generate a random number rn between 0 and 1. If rn is located in cumulative probability interval $(0, pc_{i1}]$, then the drop-off point i is allocated to picking station 1 ($Z_i = 1$) and put i into PS_1 . If rn is located in cumulative probability interval $(\sum_{j=1}^{j-1} pc_{ij}, \sum_{j=1}^j pc_{ij}]$, then the drop-off point i is allocated to picking station j ($Z_i = j$) and put i into PS_j .

Step 7. If $i < |I|$, let $i = i + 1$ go to **Step 2** until all drop-off points are assigned. If $i = |I|$, go to **Step 8**.

Step 8. Calculate the total expected number of parcels $TP = \sum_i l_i$ to be sorted by the RSS system.

Step 9. Calculate the number of AMRs allocated to each picking station j using $X_j = \lfloor r \times \frac{\sum_{i \in PS_j} l_i}{TP} \rfloor$ until all picking stations are allocated.

Step 10. Randomly select a value from the upper ($\bar{n}$) and lower ($\underline{n}$) limits of re-inspection personnel to determine the number of re-inspection personnel allocated N .

Step 11. Repeat Steps 1–10 until B initial integer solutions are derived.

4.2. Adaptive particle and Hyperball search (APHS)

In this section, we present a gradient- and convexity assumption-free algorithm, known as adaptive particle and hyperball Search, which is a modification of the discrete version developed by Chang et al. [15] to further improve the continuous solutions P_b over the feasible continuous region defined by constraint (19) under fixed integer solutions S_b , $b \in$

$\{M(r) \cup V(r-1)\}$ through the global and local search mechanisms. The overall structure of the proposed algorithm is shown in Section A2 of Appendix A. The algorithmic process starts with an initialization phase, then enters a main loop consisting of 4 steps, and concludes with a final fifth step. In the main loop of APHS, Step 1 determines the allocation of the simulation budget for each fixed integer solutions S_b . Step 2 is designed to perform a global search mechanism akin to the Particle Swarm Optimization (PSO) algorithm, referred to as APS. Step 3 is designed to further conduct a local search strategy based on Latin Hyperball Sampling [15] within the most promising region. Step 4 updates and identifies the best solution of the current iteration using OCBA-1 method. Finally, after all integer solutions have been scanned by using the first four steps respectively, Step 5 returns the best continuous solution P_b^* and the corresponding objective value $\overline{TC}(S_b, P_b^*)$ for each integer solutions S_b , $b \in \{M(r) \cup V(r-1)\}$. By combining global and local search strategies, APHS can concurrently explore the entire feasible solution space while concentrating greater sampling effort around the current best solution found so far. This dual approach enables the algorithm to efficiently and effectively identify the optimal solution. It's important to note that each complete cycle through the main loop constitutes one iteration of APHS, with the iterations indexed by the variable k .

The APHS begins by randomly generating U initial particle positions (continuous solutions), denoted as $P_{b,u,0}$, $u \in [1, U]$, for each fixed integer solutions S_b , $b \in \{M(r) \cup V(r-1)\}$. To include the original high-quality continuous solution to accelerate the convergence, we set $P_{b,1,0} = P_b$. These solutions then form the initial set of sampled solutions, referred to as H_b . Finally, for each fixed S_b , the OCBA-1 algorithm first estimates the expected objective values of U initial particle positions using the minimum number of simulations and then identify current best continuous solution P_b^* along with its corresponding optimal objective function value $\overline{TC}(S_b, P_b^*)$. These optimal objective function values will be used as the basis for allocating the number of simulation replications for different integer solutions S_b , $b \in \{M(r) \cup V(r-1)\}$, during the execution of the APHS main loop.

For each fixed integer solutions S_b , the main loop of APHS, comprising Steps 1–4, is designed to iteratively search, update, and identify the global optimum P_b^* of the continuous solution within the feasible solution space defined by constraint (19). At the start of this main loop, a specific integer solution S_b , where $b \in \{M(r) \cup V(r-1)\}$ is selected, and its allocated simulation budget/observations for searching the continuous solution are calculated using Eq. (20). An Adaptive Particle Search methodology (see Section 4.2.1), based on Particle Swarm Optimization (PSO), is employed to accelerate convergence to the globally optimal solution by moving each of the U solutions to various points within the entire feasible region. This algorithm also permits global repositioning when a particle's position fails to improve. Since particles can be repositioned to any point within the feasible region, potentially far from their original locations, the APS mechanism serves as a global search strategy that aids in converging to the global optimum. Complementing APS is a local search scheme employing Latin Hyperball Sampling (see Section 4.2.2), which significantly speeds up the search for the optimal solution. This scheme samples a user-defined number T of solutions within the most promising region. Consequently, APS facilitates exploration of the entire feasible region, while Latin Hyperball Sampling concentrates on exploiting the vicinity of the best-found solution. Finally, OCBA-1 is again employed to efficiently allocate simulation observations among the t sampled points, ensuring that, at the end of each iteration, the best solution P_b^* is identified with a certain level of confidence or within a fixed number of simulations from among the solutions explored by the algorithm so far. Table 3 also shows the detailed pseudo code of APHS.

$$obs_b = Total_sim_budget \times \frac{1/\overline{TC}(S_b, P_b^*)}{\sum_{b'=1}^{|M(r) \cup V(r-1)|} (1/\overline{TC}(S_{b'}, P_{b'}^*))} \quad (20)$$

Table 3

Pseudo code of adaptive particle and hyperball search.

Input: $U, T, P_{search}, Total_sim_budget, M(r), V(r-1)$.
Output: (S_b, P_b^*) and $\overline{TC}(S_b, P_b^*)$ for all $b \in \{M(r) \cap V(r-1)\}$.
1 Initialization: For each $b \in \{M(r) \cap V(r-1)\}$, randomly generate U initial particle positions $\{P_{b,1,0}, P_{b,2,0}, \dots, P_{b,u,0}, \dots, P_{b,U,0}\}, u \in [1, U]$ satisfying the constraint (19), under fixed S_b .
2 Let $P_{b,1,0} = P_b$ to incorporate original continuous solution P_b .
3 Estimate objective value $\overline{TC}(S_b, P_{b,u,0})$ of all U particle positions by OCBA-1 method.
4 Identify current best solution (S_b, P_b^*) from these U particles under fixed S_b .
5 Add all U particle positions $P_{b,u,0}$ into sampled set H_b .
6 For $b = 1$ to $ M(r) \cap V(r-1) $ do
7 Select the solution S_b and let $k = 1$.
8 Compute allocated simulation budget of the solution S_b by $obs_b = Total_sim_budget \times [1/\overline{TC}(S_b, P_b^*)] / [\sum_{b'=1}^{ M(r) \cap V(r-1) } (1/\overline{TC}(S_{b'}, P_{b'}^*))]$.
9 Specify the initial velocity $\mathbf{VP}_{b,u,0}$ of U particle's current positions.
10 While Accumulated simulation budget < obs_b do
11 Compute updated velocity $\mathbf{VP}_{b,u,k}$ for all U particles by Eq. (21).
12 Generate new particle solution $P_{b,u,move}$ from $P_{b,u,k-1}$ for all U particles by Eq. (22).
13 Add all U new particle positions into sampled set H_b .
14 Estimate the objective value $\overline{TC}(S_b, P_{b,u,move})$ of all particles $u = 1, \dots, U$ by OCBA-1 method.
15 For $u = 1$ to U do
16 If $\overline{TC}(S_b, P_{b,u,move}) < \overline{TC}(S_b, P_{b,u,k-1})$ and $P_{b,u,move}$ is a feasible solution satisfying the constraint (19) then
17 Update $P_{b,u,k} = P_{b,u,move}$ and let $\overline{TC}(S_b, P_{b,u,k}) = \overline{TC}(S_b, P_{b,u,move})$.
18 Else
19 Generate a random number $\rho \sim Uniform(0, 1)$.
20 If $\rho < P_{search}$ then
21 Randomly select a particle $q, q \neq u$ from current particle set.
22 Generate new particle position $P_{b,u,new}$ by uniformly sampling for each dimension d in $[\min(P_{b,u,k-1}^d, P_{b,q,k-1}^d), \max(P_{b,u,k-1}^d, P_{b,q,k-1}^d)]$.
23 In the sampling process, P_1 is first selected within the feasible range, and then P_2 is sampled within the range $[(\tau - \theta)/ P_1 \times \beta \times (1 - \theta) , 1]$.
24 Else
25 Uniformly sample new feasible particle position $P_{b,u,new}$ from entire feasible solution space defined by constraint (19).
26 In the sampling process, P_1 is first selected within the feasible range, and then P_2 is sampled within the range $[(\tau - \theta)/ P_1 \times \beta \times (1 - \theta) , 1]$.
27 End
28 Add new particle position $P_{b,u,new}$ into sampled set H_b .
29 Estimate objective value $\overline{TC}(S_b, P_{b,u,new})$.
30 If $\overline{TC}(S_b, P_{b,u,new}) < \overline{TC}(S_b, P_{b,u,k-1})$ and $P_{b,u,new}$ is a feasible solution satisfying the constraint (19) then
31 Update $P_{b,u,k} = P_{b,u,new}$ and let $\overline{TC}(S_b, P_{b,u,k}) = \overline{TC}(S_b, P_{b,u,new})$.
32 Else
33 Go back to line 19 for sampling new solution.
34 End
35 End
36 If $\overline{TC}(S_b, P_{b,u,k}) < \overline{TC}(S_b, P_b^*)$ then
37 Update current best solution (S_b, P_b^*) by $P_b^* = P_{b,u,k}$ and its best objective value $\overline{TC}(S_b, P_b^*) = \overline{TC}(S_b, P_{b,u,k})$.
38 End
39 Let $u = u + 1$.
40 End
41 Build the hyperball region ψ_k considering constraint (19) by current best solution (S_b, P_b^*) and all points in sampled set H_b .
42 Generate T feasible solutions $P_{b,t}, t \in [1, T]$, from ψ_k using Latin Hyperball Sampling and include these solutions in H_b .
43 In the sampling process, P_1 is first selected within the feasible range, and then P_2 is sampled within the range $[(\tau - \theta)/ P_1 \times \beta \times (1 - \theta) , 1]$.
44 Estimate objective value $\overline{TC}(S_b, P_{b,t})$ for each solution $(S_b, P_{b,t}), t \in [1, T]$ by OCBA-1 method.
45 Update current best solution (S_b, P_b^*) and best objective value $\overline{TC}(S_b, P_b^*)$ by $P_b^* = \underset{P \in H_b}{\operatorname{argmin}} \overline{TC}(S_b, P)$.
46 Let $k = k + 1$.
47 End
48 Let $b = b + 1$.
49 End
50 Return (S_b, P_b^*) and $\overline{TC}(S_b, P_b^*)$, for all $b \in \{M(r) \cap V(r-1)\}$.

4.2.1. Adaptive particle search (APS)

The primary objective of APS is to find solutions that improve the expected total cost by selecting a population of candidate solutions (or particles) and moving them around in the feasible solution space using a PSO-like global search and sampling mechanism. The key distinction from standard PSO is that the APS algorithm offers a new opportunity for global repositioning through a global sampling-based local search scheme when a particle's position does not improve. The APS algorithm starts by giving the current particles (solutions), with their positions $P_{b,u,k-1}$ indexed by $u \in [1, U]$, obtained from previous iteration $k-1$, a counter for APHS (macro) iterations. The velocities $\mathbf{VP}_{b,u,k-1}$ of the current particles are indexed in the same way. Meanwhile, (S_b, P_b^*) denoted the current best solution from the last iteration until now. In

each iteration, APHS iterates through each of the U particles by first updating the velocity of the current particle by Eq. (21) and then using the newly updated velocity $\mathbf{VP}_{b,u,k}$ along with the current locations $P_{b,u,k-1}$ to generate an updated solution $P_{b,u,move}$ by Eq. (22). At the same time, this updated $P_{b,u,move}$ is also added into the sampled set H_b .

$$\mathbf{VP}_{b,u,k} = \omega \times \mathbf{VP}_{b,u,k-1} + c \times (P_b^* - P_{b,u,k-1}) \times \varepsilon \quad (21)$$

$$P_{b,u,move} = P_{b,u,k-1} + \mathbf{VP}_{b,u,k} \quad (22)$$

Here, ω represents an inertia weight, c represents an acceleration constant, and ε represents a random number generated from a continuous uniform distribution in the interval $[0, 1]$.

Subsequently, using the OCBA-1 method to estimate the expected total cost $\overline{TC}(S_b, P_{b,u,move})$ of all relocated particles, a comparison is made

between $\overline{TC}(S_b, P_{b,u,move})$ and $\overline{TC}(S_b, P_{b,u,k-1})$. If the former is less than the latter and the updated particle solution $P_{b,u,move}$ also satisfies constraint (19), then the particle's current position $P_{b,u,k}$ in this iteration k is set to $P_{b,u,move}$, and its corresponding objective value $\overline{TC}(S_b, P_{b,u,k})$ is set to $\overline{TC}(S_b, P_{b,u,move})$. Moreover, if the estimated $\overline{TC}(S_b, P_{b,u,k}) < \overline{TC}(S_b, P_b^*)$, then current best solution (S_b, P_b^*) is updated by $P_b^* = P_{b,u,k}$ and the current best objective value $\overline{TC}(S_b, P_b^*)$ is updated to $\overline{TC}(S_b, P_{b,u,k})$. Finally, the u counter is advanced to the next particle among the U particles, and one iteration terminates when the counter reaches $u = U$.

However, if in the above procedure, $\overline{TC}(S_b, P_{b,u,move})$ is greater than $\overline{TC}(S_b, P_{b,u,k-1})$ or $P_{b,u,move}$ is an infeasible solution, a different mechanism is triggered to generate a new solution $P_{b,u,new}$ to update $P_{b,u,k}$. This mechanism first generates a random number ρ from the uniform distribution on the interval $[0, 1]$. If ρ is less than a pre-specified parameter P_{search} , then one of the other $U-1$ is randomly selected. Next, a new point, denoted as $P_{b,u,new}$, is generated by uniformly sampling for each dimension d , $d = 1, 2, \dots, D$, on $[\min(P_{b,u,k-1}^d, P_{b,q,k-1}^d), \max(P_{b,u,k-1}^d, P_{b,q,k-1}^d)]$. On the other hand, if ρ is greater than P_{search} , then $P_{b,u,new}$ is generated by uniformly sampling on the entire feasible region defined by constraint (19). Furthermore, during the sampling process, to satisfy constraint (19), P_1 is first selected within the feasible range, and then P_2 is sampled within the range $[(\tau - \theta)/[P_1 \times \beta \times (1 - \theta)], 1]$ based on the P_1 value obtained. At the same time, this new point $P_{b,u,new}$ is also added into the sampled set H_b . After estimating the expected total cost $\overline{TC}(S_b, P_{b,u,new})$ of this new point, a comparison is also made between $\overline{TC}(S_b, P_{b,u,new})$ and $\overline{TC}(S_b, P_{b,u,k-1})$. If the former is less than the latter and the updated particle solution $P_{b,u,new}$ also satisfies constraint (19), then the particle's current position $P_{b,u,k}$ in this iteration k is set to $P_{b,u,new}$, and its corresponding objective value $\overline{TC}(S_b, P_{b,u,k})$ is set to $\overline{TC}(S_b, P_{b,u,new})$. However, if $\overline{TC}(S_b, P_{b,u,new}) \geq \overline{TC}(S_b, P_{b,u,k-1})$ or $P_{b,u,new}$ is an infeasible solution, the algorithm returns to step 5, where a new Bernoulli trial is conducted, and a new solution is generated again. The detailed steps of the APS procedure are described as follows and shown graphically in Section A3 of Appendix A.

Step 1. Give U particle's current positions $\{P_{b,1,k-1}, \dots, P_{b,u,k-1}, \dots, P_{b,U,k-1}\}$, U particle's current velocities $\{VP_{b,1,k-1}, \dots, VP_{b,u,k-1}, \dots, VP_{b,U,k-1}\}$, and current best solution (S_b, P_b^*) from the last iteration.

Step 2. Compute the updated velocity $VP_{b,u,k}$ for all U particles by Eq. (21). Use Eq. (22) to generate new particle solution $P_{b,u,move}$ for all U particles. Add all U newly generated particle solutions into sampled set H_b . Then, estimate the objective value $\overline{TC}(S_b, P_{b,u,move})$ of all particles $u = 1, \dots, U$ through the OCBA-1 method. Let $u = 1$.

Step 3. If $\overline{TC}(S_b, P_{b,u,move}) < \overline{TC}(S_b, P_{b,u,k-1})$ and $P_{b,u,move}$ is a feasible solution satisfying the constraint (19), go to Step 4. If $\overline{TC}(S_b, P_{b,u,move}) \geq \overline{TC}(S_b, P_{b,u,k-1})$ or $P_{b,u,move}$ is an infeasible solution, go to Step 5.

Step 4. Update particle u 's position by $P_{b,u,k} = P_{b,u,move}$ and let $\overline{TC}(S_b, P_{b,u,k}) = \overline{TC}(S_b, P_{b,u,move})$. Go to Step 8.

Step 5. Generate a random number ρ from $Uniform(0, 1)$. If $\rho < P_{search}$, randomly select a particle q , $q \neq u$ from the current particle set. Employ uniformly sample for each dimension d , $d = 1, 2, \dots, D$, on $[\min(P_{b,u,k-1}^d, P_{b,q,k-1}^d), \max(P_{b,u,k-1}^d, P_{b,q,k-1}^d)]$ as $P_{b,u,new}$. Else, uniformly sample a feasible solution $P_{b,u,new}$ from the entire feasible solution space defined by constraint (19). In the sampling process, P_1 is first selected within the feasible range, and then P_2 is sampled within the range $[(\tau - \theta)/[P_1 \times \beta \times (1 - \theta)], 1]$ based on the P_1 value obtained. Add this new particle position $P_{b,u,new}$ to the sampled set H_b , and estimate its objective value $\overline{TC}(S_b, P_{b,u,new})$. Go to Step 6.

Step 6. If $\overline{TC}(S_b, P_{b,u,new}) < \overline{TC}(S_b, P_{b,u,k-1})$ and $P_{b,u,new}$ is a feasible

solution satisfying the constraint (19), then go to Step 7. If $\overline{TC}(S_b, P_{b,u,new}) \geq \overline{TC}(S_b, P_{b,u,k-1})$ or $P_{b,u,new}$ is an infeasible solution, go back to Step 5 for sampling new solution.

Step 7. Update particle u 's position by $P_{b,u,k} = P_{b,u,new}$ and let $\overline{TC}(S_b, P_{b,u,k}) = \overline{TC}(S_b, P_{b,u,new})$. Go to Step 8.

Step 8. If $\overline{TC}(S_b, P_{b,u,k}) < \overline{TC}(S_b, P_b^*)$, update current best solution (S_b, P_b^*) by $P_b^* = P_{b,u,k}$ and its best objective value $\overline{TC}(S_b, P_b^*) = \overline{TC}(S_b, P_{b,u,k})$. Let $u = u + 1$. If $u > U$, go to Step 9. Else, go to Step 3.

Step 9. Return (S_b, P_b^*) , $\overline{TC}(S_b, P_b^*)$, and H_b .

4.2.2. Latin Hyperball sampling (LHS)

After APS generates U new particle solutions and potentially updates the current best solution P_b^* , the next step involves defining the most promising region and generating T new solutions within this feasible region using Latin-Hyperball-Sampling-based local search mechanism, thus allowing APHS to focus on the vicinity of the best-found solution and greatly enhance the algorithm's efficiency through this exploitative procedure. Consequently, the LHS-based local search algorithm comprises two main components: (1) identifying the most promising region by constructing hyperball neighborhoods of all sampled points (steps 1 to 7), and (2) generating new solutions within the most promising region to further update the current best solution (steps 8 to 11).

In the first part of LHS, because the most promising region ψ_k is a hyperball centered on the current best solution P_b^* returned by APS, it must be determined by constructing the hyperball neighborhoods of all points in the sampled set H_b to identify the extent of the most promising region ψ_k . To define all of the neighborhoods in H_b , we define two sets Λ and Γ . Initially, Λ contains all the points in H_b and is reduced by one point in each iteration until it becomes empty. Conversely, Γ starts as an empty set and is incremented by one point in each iteration until it includes all the points in H_b . Crucially, a certain point $P_{b,v}$ is moved from Λ to Γ if it is randomly chosen from Λ and its neighborhood is defined in that iteration. The neighborhood of $P_{b,v}$ is determined based on whether its nearest Euclidean neighbor $P_{b,v'}$ has already been added to Γ and thus had its neighborhood previously defined. If the neighborhood of $P_{b,v}$ has not been defined yet, the neighborhood of $P_{b,v}$ is defined as a hyperball with a radius equal to half the Euclidean distance $Dist(P_{b,v}, P_{b,v'})$ (see Eq. (23)) between $P_{b,v}$ and $P_{b,v'}$, i.e. $rd_v = \frac{1}{2} \{Dist(P_{b,v}, P_{b,v'})\}$. If, however, $P_{b,v'}$'s neighborhood has already been established, we can calculate the minimum distance between $P_{b,v}$ and all previously defined neighborhoods using $z = \min\{Dist(P_{b,v}, P_{b,v'}) - rd_{v'} | P_{b,v'} \in \Gamma\}$ and compute the neighborhood hyperball radius of $P_{b,v}$ to be $rd_v = \min\{z, Dist(P_{b,v}, P_{b,v'}) - rd_{v'}\}$, where $rd_{v'}$ is defined as the neighborhood radius of $P_{b,v'}$. At the end of each iteration, the hyperball neighborhood region of $P_{b,v}$ can be represented by Eq. (24). After applying this procedure to obtain the hyperball neighborhoods of all points, we can identify the most promising region ψ_k by considering the intersection between the feasible solution region defined by constraint (19) and the neighborhood region $\delta(P_b^*)$ around the current best solution P_b^* . Fig. 3 illustrates the resulting neighborhoods after applying this procedure. In the example shown, there are eight solutions in total, with $P_{b,4} = P_b^*$ identified as the current best solution. Thus, the region surrounding $P_{b,4}$ is marked as the most promising region.

$$Dist(P_{b,v}, P_{b,v'}) = \sqrt{(P_{1,b,v} - P_{1,b,v'})^2 + (P_{2,b,v} - P_{2,b,v'})^2} \quad (23)$$

$$\delta(P_{b,v}) = \{P_{b,v'} \in \Theta | Dist(P_{b,v}, P_{b,v'}) \leq rd_v\} \quad (24)$$

The second phase of LHS focuses on identifying T solutions within ψ_k by creating a near-random sample of parameter values, which is crucial for boosting the sampling efficiency in the proposed algorithm. Given that ψ_k is a hyperball, we employ a method known as *Latin Hyperball*

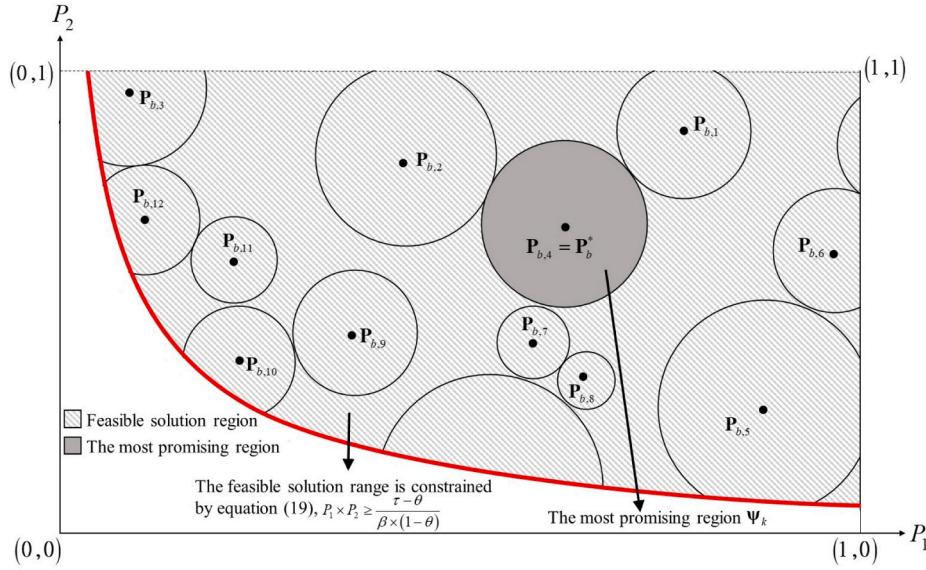


Fig. 3. Partitioning the feasible region into neighborhoods and identifying the most promising one.

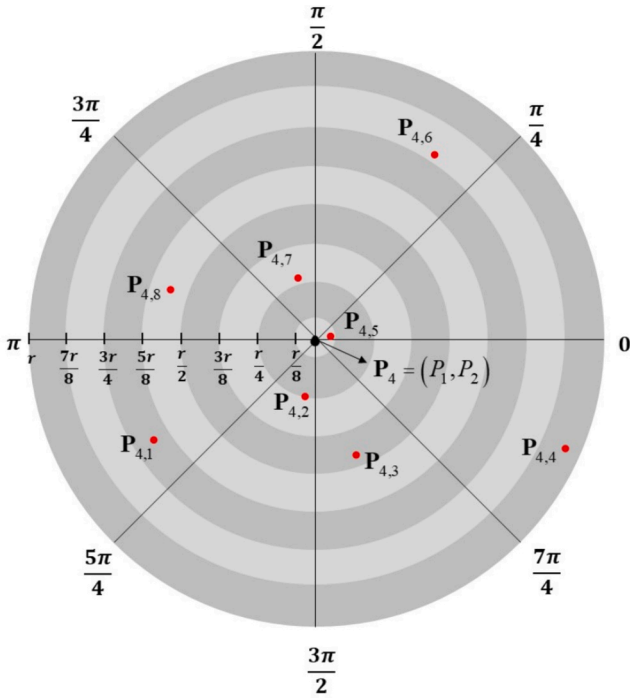


Fig. 4. Illustration of Latin Hyperball Sampling with the selection of $T = 8$ in the most promising 2-Dimensional ψ_k .

Sampling to generate a diverse set of solutions within this region ψ_k . Latin Hyperball Sampling, which can be seen as a hypersphere adaptation of Latin Hypercube Sampling [25], allows for more efficient exploration of the feasible region. Unlike random uniform sampling, it ensures even distribution over the solution space with a smaller sample size. As a result, Latin Hyperball Sampling is applied in the most promising region to enhance the convergence rate of APhS. The Latin Hyperball Sampling scheme of generating a specific sample point $P_{b,t}$, where $t \in [1, T]$, from the most promising region ψ_k is summarized with the following Eq. (25) and brief explanation:

$$P_{b,t} = \left(o_1 + rc_t \prod_{d=1}^{D-1} \cos \theta_{td}, o_2 + rc_t \prod_{d=2}^{D-1} \cos \theta_{td} \sin \theta_{t1}, \dots, o_{D-1} + rc_t \prod_{d=D-1}^{D-1} \cos \theta_{td} \sin \theta_{t(D-2)}, o_D + rc_t \sin \theta_{t(D-1)} \right) \quad (25)$$

In this context, t ranges from 1 to T , as the Eq. (25) is used to select T solutions for sampling within ψ_k . Each of these T solutions has D components, reflecting the D -dimensional nature of the feasible region. In the expression above, the center of the most promising region ψ_k is defined by the coordinates $(o_1, o_2, \dots, o_D)$. The term rc_t represents one realization of a continuous uniform random variable, where the parameters correspond to a randomly chosen value from a set of T equally Euclidean increments of the radius of ψ_k . Similarly, each θ term is chosen from a continuous uniform distribution, where the parameters represent a randomly selected value from a set of T equal radian increments of the total number of “radians” in the D -sphere. Fig. 4 illustrates the application of Latin Hyperball Sampling for selecting $T = 8$ points to sample within a 2-dimensional region ψ_k . In this illustration, $P_4 = P_b^*$ indicates the center of the most promising region, while $\{P_{4,1}, P_{4,2}, \dots, P_{4,8}\}$ represent the eight points selected for sampling by the Latin Hyperball Sampling method. After the T solutions are chosen through LHS from the most promising region ψ_k , these solutions are added to the sampled set H_b . Finally, the OCBA-1 method is employed to estimate the objective value $\overline{TC}(S_b, P_{b,t})$ of each solution $(S_b, P_{b,t})$, $t \in [1, T]$ and current best solution (S_b, P_b^*) and best objective value $\overline{TC}(S_b, P_b^*)$ are updated by $P_b^* = \underset{P \in H_b}{\operatorname{argmin}} \overline{TC}(S_b, P)$. The detailed LHS-based local search procedure is described as follows.

Step 1. Give current best solution (S_b, P_b^*) and all points in the sampled set H_b . Create two empty sets Λ and Γ , respectively. Then make a copy of the set H_b into set Λ . The empty set Γ will be used to store the set of already defined neighboring points.

Step 2. Calculate all Euclidean distances between all pairs of points in the sampled set H_b by using Eq. (23).

Step 3. Randomly select a point $P_{b,v}$ from Λ and then identify its nearest point, denoted by $P_{b,v'}$, in terms of Euclidean distance obtained from Step 2. Define the radius of $P_{b,v}$ as rd_v .

Step 4. If $P_{b,v} \notin \Gamma$, then calculate the neighborhood radius of $P_{b,v}$ to be $rd_v = \frac{1}{2} \{ \operatorname{Dist}(P_{b,v}, P_{b,v'}) \}$. Otherwise, Calculate the minimum

distance between $\mathbf{P}_{b,v}$ and all previously defined neighborhoods using $z = \min\{\text{Dist}(\mathbf{P}_{b,v}, \mathbf{P}_{b,v'}) - rd_{v'} | \mathbf{P}_{b,v'} \in \Gamma\}$ and the neighborhood radius of $\mathbf{P}_{b,v}$ to be $rd_v = \min\{z, \text{Dist}(\mathbf{P}_{b,v}, \mathbf{P}_{b,v'}) - rd_{v'}\}$.

Step 5. Determine the hyperball neighborhood region $\delta(\mathbf{P}_{b,v})$ of $\mathbf{P}_{b,v}$ by using Eq. (24). Delete $\mathbf{P}_{b,v}$ from Λ and insert $\mathbf{P}_{b,v}$ into Γ .

Step 6. If the set Λ is empty, the neighborhood region of all points in the sampled set $\mathbf{H}_b$ is constructed and then go to **Step 7**. Otherwise, go back to **Step 3**.

Step 7. Identify the most promising region ψ_k by considering the intersection between the feasible solution region defined by constraint (19) and the neighborhood region $\delta(\mathbf{P}_b^*)$ around the current best solution $\mathbf{P}_b^*$.

Step 8. Employ the Latin Hyperball Sampling to user-defined number T of solutions, $\{\mathbf{P}_{b,1}, \mathbf{P}_{b,2}, \dots, \mathbf{P}_{b,t}, \dots, \mathbf{P}_{b,T}\}$, from the most promising region by using Eq. (25). In the sampling process, P_1 is first selected within the feasible range, and then P_2 is sampled within the range $[(\tau - \theta)/[P_1 \times \beta \times (1 - \theta)], 1]$ based on the P_1 value obtained.

Step 9. After the T solutions are chosen through LHS from the most promising region ψ_k , these solutions are added to the sampled set $\mathbf{H}_b$.

Step 10. Estimate the objective value $\overline{TC}(\mathbf{S}_b, \mathbf{P}_{b,t})$ of each solution $(\mathbf{S}_b, \mathbf{P}_{b,t})$, $t \in [1, T]$ through the OCBA-1 method.

Step 11. Update and return current best solution $(\mathbf{S}_b, \mathbf{P}_b^*)$ and best objective value $\overline{TC}(\mathbf{S}_b, \mathbf{P}_b^*)$ by $\mathbf{P}_b^* = \underset{\mathbf{P} \in \mathbf{H}_b}{\text{argmin}} \overline{TC}(\mathbf{S}_b, \mathbf{P})$.

4.3. OCBA- m based objective screening

In this section, we propose the OCBA- m based screening algorithm [19] to effectively screen out the top m subsets with superior objective values from the tactical design solution set $\{M(r) \cup V(r-1)\}$. This selection, aimed at forming the next generation of surviving solutions $V(r)$, is based on the estimated expected objective values $\overline{TC}(\mathbf{S}_e, \mathbf{P}_e^*)$ obtained from the RSS simulation model with multiple replications. However, evaluating the objective values through the simulation model requires a large number of simulation runs and spends a significant amount of computation time. Therefore, the primary goal of the OCBA- m based screening algorithm is to maximize the probability of correctly (PCS) for identifying the top m systems from a set of k systems within a fixed computational simulation budget to reduce the total computation time. It intelligently allocates this limited simulation budget across different tactical design solutions and sequentially determines which tactical design alternative requires more simulation observations/replications, as well as the number of additional observations/replications needed to achieve the maximization of PCS $P\{CS\}$. The OCBA-1 method is merely a special case of the OCBA- m algorithm when $m = 1$. The detailed OCBA- m based objective screening procedure is described as follows.

Step 1. Specify $E = |M(r) \cup V(r-1)|$ tactical design solutions obtained from previous APHS stage, initial simulation replication n_0 , total simulation replications T , additional number of simulation replications in each iteration Δ , predefined acceptable $P\{CS\}_{desired}$ and let $l = 0$.

Step 2. Execute n_0 simulation replications in the RSS simulation model for all tactical design solutions $(\mathbf{S}_e, \mathbf{P}_e^*)$, where $e \in \{M(r) \cup V(r-1)\}$; $N_1^l = N_2^l = \dots = N_E^l = n_0$.

Step 3. Calculate sample means $\bar{J}_e = \overline{TC}(\mathbf{S}_e, \mathbf{P}_e^*)$ (see Eq. (26)), sample standard deviations δ_e (see Eq. (27)) and value σ_e (see Eq. (28)) for each tactical design solution $(\mathbf{S}_e, \mathbf{P}_e^*)$, $e \in \{M(r) \cup V(r-1)\}$, using simulation output performance with N_e^l simulation replications.

$$\bar{J}_e = \overline{TC}(\mathbf{S}_e, \mathbf{P}_e^*) = \sum_{z=1}^{N_e^l} TC(\mathbf{S}_e, \mathbf{P}_e^*, \omega^{(z)}), \quad e \in \{M(r) \cup V(r-1)\} \quad (26)$$

$$\delta_e = \sqrt{\frac{1}{N_e^l - 1} \sum_{z=1}^{N_e^l} [TC(\mathbf{S}_e, \mathbf{P}_e^*, \omega^{(z)}) - \overline{TC}(\mathbf{S}_e, \mathbf{P}_e^*)]^2}, \quad e \in \{M(r) \cup V(r-1)\} \quad (27)$$

$$\sigma_e = \frac{\delta_e}{\sqrt{N_e^l}} \quad (28)$$

Step 4. Sort tactical design solution indexes in ascending order based on the value of $\bar{J}_e$ (i.e. $\bar{J}_{(1)} \leq \bar{J}_{(2)} \leq \dots \leq \bar{J}_{(E)}$). Based on this ranking, select the top m tactical design solutions and place them into the subset SS_m .

Step 5. Based on the ranking in **Step 4**, select the sample means, $\bar{J}_{(m)}$ and $\bar{J}_{(m+1)}$, and its corresponding values, $\sigma_{(m)}$ and $\sigma_{(m+1)}$, of the m -th and $(m+1)$ -th ranked samples to calculate the constant value c by Eq. (29).

$$c = \frac{\sigma_{(m+1)} \times \bar{J}_{(m)} + \sigma_{(m)} \times \bar{J}_{(m+1)}}{\sigma_{(m)} + \sigma_{(m+1)}} \quad (29)$$

Step 6. Based on the constant value c obtained **Step 5**, calculate the value $\lambda_e = \bar{J}_e - c$ for each tactical design solution $(\mathbf{S}_e, \mathbf{P}_e^*)$, $e \in \{M(r) \cup V(r-1)\}$.

Step 7. Calculate $P\{CS\}$ by using Eq. (30). Then, if $P\{CS\} \geq P\{CS\}_{desired}$ or $\sum_{e=1}^{|M(r) \cup V(r-1)|} N_e^l \geq T$, stop and go to **Step 10**. Otherwise, go to **Step 8**.

$$P\{CS\} = \prod_{e \in SS_m} \Phi\left(\frac{c - \bar{J}_e}{\sqrt{\frac{\delta_e^2}{N_e^l}}}\right) \times \prod_{e \notin SS_m} \Phi\left(\frac{\bar{J}_e - c}{\sqrt{\frac{\delta_e^2}{N_e^l}}}\right) \quad (30)$$

Step 8. Increase the additional number of simulation budget by Δ , and calculate the new simulation budget allocation, $N_1^{l+1}, N_2^{l+1}, \dots, N_E^{l+1}$, of all tactical design solutions $(\mathbf{S}_e, \mathbf{P}_e^*)$, where $e \in \{M(r) \cup V(r-1)\}$ according to the following Eq. (31).

$$\frac{N_1^{l+1}}{\left(\frac{\delta_1}{\lambda_1}\right)^2} = \frac{N_2^{l+1}}{\left(\frac{\delta_2}{\lambda_2}\right)^2} = \dots = \frac{N_E^{l+1}}{\left(\frac{\delta_E}{\lambda_E}\right)^2} \quad (31)$$

Step 9. Perform additional $\max(N_e^{l+1} - N_e^l, 0)$ simulations for each tactical design $(\mathbf{S}_e, \mathbf{P}_e^*)$, $e \in \{M(r) \cup V(r-1)\}$. Then, let $l = l+1$ and go back to **Step 3**.

Step 10. Return m surviving tactical design solutions $(\mathbf{S}_e, \mathbf{P}_e^*)$ in subset SS_m and their corresponding objective value $\overline{TC}(\mathbf{S}_e, \mathbf{P}_e^*)$ and put their corresponding index into the set of surviving solutions $V(r)$.

4.4. Generation of the candidate solutions

This algorithm introduces an innovative strategy for generating candidate solutions, utilizing a replenishment probability RP , to effectively balance and enhance both exploitation and exploration within the solution space. To enhance exploitation, it makes sense to generate new candidate solutions with a replenishment probability RP from the neighborhoods of the surviving solutions $V(r-1)$, as these solutions are likely to be promising. For the given drop-off assignment surviving solution $\mathbf{Z}$, we develop the following four rules to generate new candidate solutions. **Rule (1)** first selects pairs of adjacent elements in the vector-form drop-off assignment solution where the values differ by more than γ . This assignment solution is then split into left and right sections. If the selected adjacent pair is in the left half, the larger value in the pair is replaced by the smaller value. If the pair is in the right half, the smaller value is replaced by the larger value. For example, given the surviving solution 314212232411324, applying Rule (1) with $\gamma = 2$ results in the

candidate solution 112212234443344. **Rule (2)** generates new candidate solutions by sorting the values in the vector-form drop-off assignment solution in ascending order. For example, given the surviving solution 314212232411324, applying Rule (2) results in the candidate solution 111122222333444. **Rule (3)** partitions the vector-form drop-off assignment solution into different segments of length α . Within each segment, the values are sorted in ascending order to generate new candidate solutions. For example, given the surviving solution 314212232411324, applying Rule (3) with $\alpha = 5$ results in the candidate solution 112342223411234. **Rule (4)** first identifies pairs of adjacent elements in the vector-form drop-off assignment solution where the values differ by 1 and then randomly selects β pairs from these and swaps the values within each selected pair. For example, given the surviving solution 314212232411324, applying Rule (4) with $\beta = 3$ results in the candidate solution 112221243434344. For the given AMRs allocation surviving solution $\mathbf{X}$, we define a range of $\pm\eta$ for each element in the vector-form AMRs allocation solution to form a hypercube, and then randomly sample within this hypercube to generate new candidate solutions. For example, given the surviving solution 6324, with a hypercube $\{[4, 8], [1, 5], [0, 4], [2, 6]\}$ of $\eta = 2$, random sampling within this hypercube produces a candidate solution such as 7567. For the given re-inspection personnel allocation surviving solution N , we define a range of $\pm\pi$ around the value N , and then randomly sample within this range to generate new feasible candidate solutions. On the other hand, to facilitate exploration, there is a $1 - RP$ probability of generating new candidate solutions from the global feasible space by applying the greedy heuristic initialization method developed in Section 4.1. The candidate solutions are continuously produced in each iteration until the set $M(r)$ is full. The steps for generating candidate solutions are summarized below.

Step 1. Set neighborhood set $M(r) = \phi$ and $b = 1$.

Step 2. Generate a random number $\kappa = \text{Uniform}(0, 1)$. Then, if $\kappa < RP$, go to **Step 3**. Otherwise, go to **Step 7**.

Step 3. (Exploitation) Randomly select a specific surviving solution $\{Z, \mathbf{X}, N\}$ from survival set $V(r - 1)$.

Step 4. (Exploitation) For the given drop-off assignment surviving solution Z , randomly choose one of four rules, **Rule (1)**, **Rule (2)**, **Rule (3)** and **Rule (4)**, to generate new candidate solution Z' .

Step 5. (Exploitation) For the given AMRs allocation surviving solution $\mathbf{X}$, we define a range of $\pm\eta$ for each element in the vector-form AMRs allocation solution to form a hypercube, and then randomly sample within this hypercube to generate new candidate solution $\mathbf{X}'$.

Step 6. (Exploitation) For the given re-inspection personnel allocation surviving solution N , we define a range of $\pm\pi$ around the value N , and then randomly sample within this range to generate new feasible candidate solution N' . go to **Step 8**.

Step 7. (Exploration) Generate new candidate solution $(Z', \mathbf{X}', N')$ from the global feasible space by applying the greedy heuristic initialization method developed in Section 4.1. Go to **Step 8**.

Step 8. Add the index b of this newly generated candidate solution $(Z', \mathbf{X}', N')$ into $M(r)$. Update $b = b + 1$. If $b > B$, end the procedure and return $M(r)$ and all newly generated candidate solutions. Otherwise, go back to **Step 2**.

5. Empirical study

This section presents an empirical study conducted in collaboration with Chunghwa Post Co., Ltd. in Taiwan, applying the mixed-integer simulation optimization model and the ORS-APHS algorithm proposed in this study to the tactical design decisions of the robotic sorting system for international inbound parcel processes at a newly established mail logistics center. This empirical study demonstrates the computational performance of the proposed ORS-APHS algorithm and uses sensitivity analysis to provide management and design insights for designers of the robotic sorting system with guaranteed safety levels for international

inbound parcel processes. Section 5.1 details the components of the robotic sorting system for international inbound parcel processes, the parameters for constructing the discrete event simulation model, and presents the results of the optimized tactical design decisions, including drop-off point assignment, AMRs allocation, and re-inspection personnel allocation solutions. Section 5.2 performs several experiments to showcase the effectiveness and efficiency of the proposed ORS-APHS algorithm. Section 5.3 explores the sensitivity analysis on four parameters and discusses their managerial implications: system security levels, the number of available AMRs, physical re-inspection times, and different drop-off point distributions for arriving parcels. All numerical experiments are performed on a computer with an Intel Core i7-8665U 2.1 GHz processor and 32 GB of RAM. The RSS system's discrete event simulation is built using Flexsim simulation software, while the ORS-APHS algorithm is implemented in Python language.

5.1. Case study on the international inbound parcel operations of Taiwan postal service

Due to the growth of e-commerce combined with free trade, many consumers purchase foreign goods online, which are then delivered to their homes through international mail or parcels via the postal service logistics system. This not only significantly increases the usage of the postal logistics system but also exposes security vulnerabilities due to outdated infrastructure [65]. Particularly in Taiwan, a country with frequent e-commerce interactions with China, anyone can send illicit goods (e.g. counterfeit personal protective equipment (PPE), counterfeit medication, illicit narcotics etc.) to Taiwan from China via mail, posing significant national security and health risks. Therefore, now more than ever, there is a crucial need to significantly enhance Chunghwa Post's efficiency in inspecting suspicious parcels. By establishing a new mail logistics processing center, Chunghwa Post has revisited the international import parcel workflow and integrated a new robotic sorting system with guaranteed safety level to enhance the sorting efficiency of international import parcels and the inspection efficiency of suspicious parcels to prevent the influx of illicit goods into the country.

Consequently, this section, through collaboration with Chunghwa Post in Taiwan, considers their implemented robotic sorting system with guaranteed safety level for postal international parcel service as the target system and applies the proposed ORS-APHS algorithm to optimize the tactical design decisions of the RSS system for the computational study. Fig. 5 illustrates that the real layout of the robotic parcel sorting system from Chunghwa Post company follows an "E-shaped" structure. [55]. This robotic parcel sorting system includes 4 picking stations and 146 drop-off points. Each drop-off point represents the destination where international parcels are sent after being sorted by the RSS system. These destinations include local postal offices for parcel delivery and other larger mail logistics centers for further detailed sorting to other locations. There are a total of 34 AMRs assigned across the picking stations to perform parcel transport and sorting operations. Each picking station has specified parking space limitations for AMRs waiting to load parcels; for example, the area in front of the first picking station can accommodate up to 7 AMRs, while the second can accommodate up to 10 AMRs. Additionally, to prevent workload imbalances among the picking stations, each station has a maximum number of drop-off points it can serve, such as the first picking station, which can serve up to 50 drop-off points. The initial inspection station of this robotic parcel sorting system uses X-ray machines for preliminary scanning inspections. Parcels suspected of containing illegal items are forwarded to a second station where re-inspection customs personnel perform physical unpacking inspections, with a maximum of 15 inspectors working simultaneously. Approximately 40 trucks deliver an average of 27,000 international import parcels daily to the mail logistics center for sorting operations. The parameters related to the robotic parcel inspection system for international postal service are outlined in Table 4.

To apply mixed-integer simulation optimization algorithms for

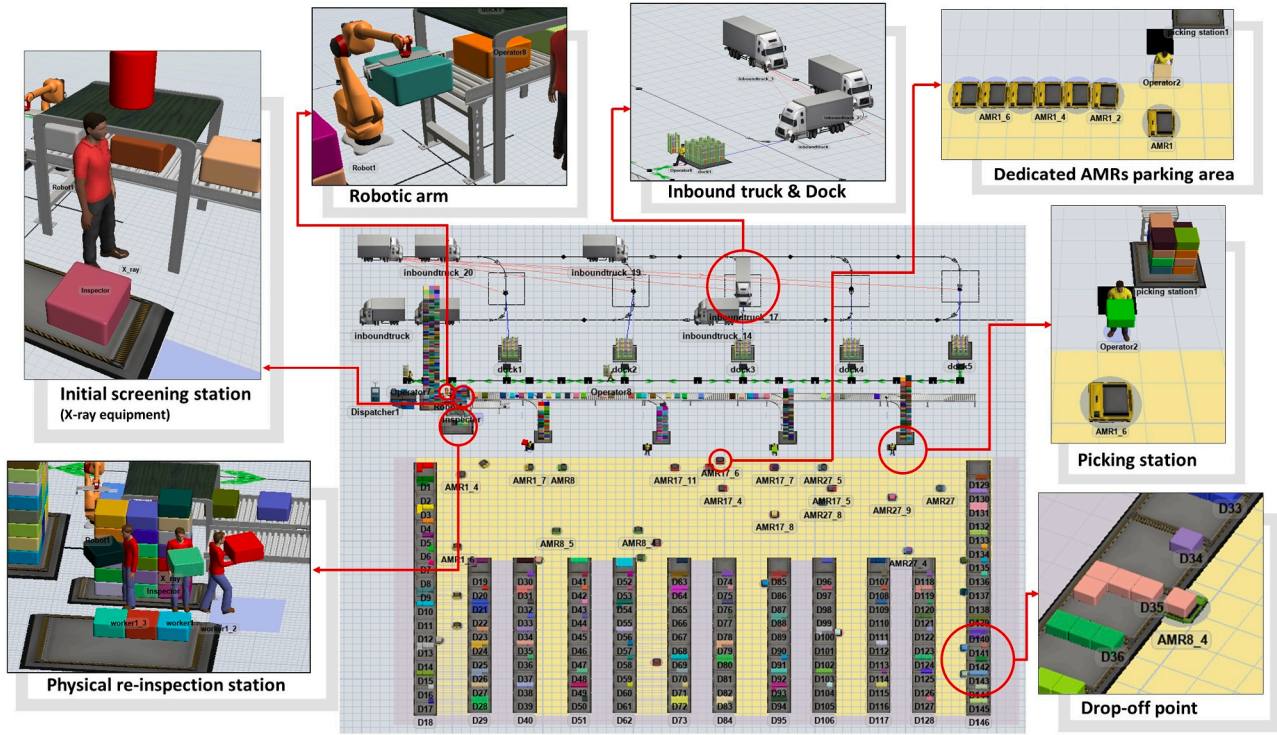


Fig. 5. Snapshot of the high-resolution simulation model depicting the real layout of the robotic parcel sorting system from Chunghwa Post company.

solving the tactical design decisions of the RSS system, we first use Flexsim simulation software to create a discrete event simulation of the RSS system. This simulation system focuses solely on the inbound operations of the international inbound parcel sorting process. It covers the scope from when trucks deliver international import parcels to the mail logistics center's dock until all parcels are sorted by AMRs and delivered to their respective drop-off points. To consider the stochastic and dynamic nature of this RSS system, many related parameters required for the simulation model, including truck inter-arrival time, number of baskets loaded on each truck, number of parcels in each basket, inspection time at initial screening and physical re-inspection stations, AMR loading and unloading time, are modeled using specific probability distributions. We collect historical data from the postal information system of Chunghwa Post company and the customs information system and use statistical software to identify the most suitable probability distribution for each parameter. Furthermore, each incoming international parcel inherently has a specific sorting destination, which is a particular drop-off point. We also collect data on the distribution of drop-off points for parcels that arrived over the past year and represent this using a Multinoulli distribution to model the probability distribution of drop-off points for incoming parcels in the simulation model. Fig. A3 in Section A3 of Appendix A shows a schematic of the Multinoulli distribution for the distribution of drop-off points for arriving parcels. Additionally, analysis of historical data has revealed that the re-inspection time of personnel at physical re-inspection stations is significantly influenced by the thoroughness or strictness of the parcel inspection, known as P_2 . The stricter or more thorough the re-inspection, the longer the time required. Therefore, after discussions with practical decision-makers, a linear relationship, $20 \times P_2 + 10$, between re-inspection time and inspection thoroughness is established to represent this phenomenon. Fig. 6 shows a snapshot of the discrete event simulation model of the RSS system and Table 3 organizes the relevant parameters required for this simulation model.

Using the proposed ORS-APHS algorithm to meet the goal of minimizing total costs within a system security level constraint of 0.0009 (θ), the RSS system sets the proportion of parcels selected for thorough

physical re-inspection at 9 % (P_1) after initial screening, with physical re-inspection stations required to operate at 100 % (P_2) strictness to inspect for illegal parcels. The RSS system requires a total of 6 re-inspection personnel (N), with each picking station needing 7 AMRs (X_1), 8 AMRs (X_2), 11 AMRs (X_3), and 8 AMRs (X_4) respectively. Fig. 7 shows the assignment results (Y_{ij}) of each drop-off point to various picking stations. Drop-off points marked with identical colors denote assignments to the same picking station. The total expected cost required for this tactical design decision is approximately 38,191 NTD. Fig. 6 also illustrates the optimal tactical design of the robotic sorting system with guaranteed security levels for the case study. Based on these results, for the Chunghwa Post company, considering the current total volume of international import parcels, the operational efficiency of the RSS system can be effectively enhanced, and operational costs reduced, while maintaining a guaranteed security level. This can be achieved by efficiently allocating the required number of AMRs to each picking station and, as shown in Fig. 6, evenly and fairly distributing the locations and quantities of drop-off points assigned to each picking station (ensuring each drop-off point is allocated to the nearest picking station). In addition, to ensure a guaranteed level of security in the RSS system, physical re-inspection stations must be staffed with sufficient personnel ($N = 6$) who are rigorously trained to achieve 100 % detection ($P_2 = 100\%$) of illegal parcels, thereby effectively combating illicit trade.

5.2. Performance analysis of ORS-APHS algorithm

For the case study of a robotic sorting system used in international inbound parcel processing, this section demonstrates the efficiency and effectiveness of the proposed ORS-APHS algorithm through extensive numerical experiments. Section 5.2.1 investigates the key parameters that significantly affect the performance of the ORS-APHS algorithm and simultaneously determines the optimal parameter settings for this algorithm using the Taguchi method. Section 5.2.2 explores the superiority of the innovative heuristic initial solution generation method and neighborhood solution generation rules introduced in the ORS-APHS algorithm by comparing with random initialization and neighborhood

Table 4
Relevant parameters values for the case study.

Parameters	Value or distribution	
Parameters of robotic sorting system for postal international parcel service	Number of drop-off points	146
	Number of picking stations	4
	Maximum number of drop-off points served by picking station (s_j)	[50, 50, 50, 50]
	Maximum number of AMRs allowed to be parked at picking station (a_i)	[7,10,11,9]
	Setup or acquisition cost of one AMR (c_1)	200,000
	Personnel cost of one re-inspector (c_2)	1,360
	Operational cost per unit of time (c_3)	1
	Maximum number of AMRs that can be used in the RSS system (r)	34
	Maximum number of personnel at the thorough physical re-inspection station ($\bar{n}$)	15
	Minimum number of personnel at the thorough physical re-inspection station ($\underline{n}$)	1
	Prespecified threshold of the system security level (θ)	0.0009
	Probability of illegal parcels being a threatening risk ($P\{T\} = \tau$)	0.001
	Probability that the illegal parcels selected to the physical re-inspection station through initial screening at stage 1 indeed pose a threat ($P\{T S_{1d}\} = \beta$)	0.0011
Parameters of discrete event simulation of robotic sorting system for postal international parcel service	Total number of truck arrivals (1/day)	40
	Truck inter-arrival time (sec)	Exponential distribution (500)
	Number of baskets loaded on each truck	Normal distribution (15,2)
	Number of parcels in each basket	Normal distribution (45,10)
	Drop-off point distribution for arriving parcels	Multinoulli distribution based on historical data (see Section A3 of Appendix A)
	Inspection time of X-ray equipment at initial screening station (sec)	Exponential distribution (3)
	Re-inspection time of each re-inspection personnel at physical re-inspection station (sec)	Exponential distribution ($20 \times P_2 + 10$)
	The time for the robotic arm to move a parcel to the conveyor belt (sec)	Exponential distribution (0.5)
	The time spent moving parcels at the picking station (sec)	Normal distribution (1,0.5)
	AMR loading and unloading time for each parcel (sec)	Normal distribution (2,0.5)
	AMR travel speed (m/sec)	2
	Conveyor belt speed (m/sec)	2

generation strategies. Section 5.2.3 shows the superior performance of the ORS-APHS algorithm compared to other established algorithms.

5.2.1. Parameter analysis by the Taguchi method

To identify the key parameters affecting the performance of the ORS-APHS algorithm and their optimal configurations, this section conducts a two-level Taguchi method-based experimental design focusing on the seven most important parameters of the algorithm. These seven parameters include the initial number of simulations in OCBA- m (n_0), maximum number of solutions of each iteration (B), number of particle positions in APHS (U), total simulation budgets (TS), number of surviving solutions (m), number of samples drawn by Latin Hyperball Sampling (T), and additional number of simulation replications in each iteration in OCBA- m (Δ). The high and low levels of each parameter are shown in Fig. 8. Using an L8 (2^7) orthogonal array design and after conducting 30 repeated experiments for each parameter combination, we obtained the main effect plots for the Signal-to-noise (S/N) ratio and the significance p-value for each parameter, as illustrated in Fig. 7. From the results in Fig. 8, number of particle positions U , total simulation budgets TS , number of surviving solutions m , number of samples drawn by LHS T are most significant and critical parameters of the ORS-APHS algorithm under $\alpha = 0.05$. The initial number of simulations in OCBA- m and maximum number of solutions of each iteration have a slight influence on the objective value. The best parameter settings of n_0 , B , U , TS , m , T and Δ are 5, 20, 5, 15,000, 2, 8 and 10, respectively. These optimal parameters are utilized in the following experimental procedures.

5.2.2. Comparison among various initialization and neighborhood strategies adopted by ORS-APHS

In this section, we undertake a numerical experiment to assess how the novel heuristic initial solution generation and heuristic neighborhood solution creation rules, introduced in the ORS-APHS algorithm of this study, enhance overall performance. Initial solution generation strategies can be categorized into “random initialization generation method” and “greedy heuristic initialization method” (see Section 4.1), while neighborhood solution creation rules can also be divided into “random neighborhood creation rule” and “innovative heuristic neighborhood creation rule” (see Section 4.4). Therefore, we examine the relative performance of the following four combinations: (1) ORS-APHS with total random strategies (referred to as “Total Random Strategies”), (2) ORS-APHS considering only the random neighborhood creation rule (referred to as “Random Neighborhood Strategy”), (3) ORS-APHS considering only the random initialization generation method (referred to as “Random Initialization Strategy”), and (4) the ORS-APHS proposed in this study, which incorporates both the greedy heuristic initialization method and the innovative heuristic neighborhood creation rule.

Due to the stochastic nature of the algorithms, we conducted 30 repeated runs with different random seeds for each of the four combinations, calculated their average objective values, and displayed their box plots in Fig. 8. Fig. 8 demonstrates that the ORS-APHS algorithm, which incorporates both the greedy heuristic initialization method and the innovative heuristic neighborhood creation rule, significantly outperforms the other three combinations in terms of average total cost and variance across 30 repeated runs with different random seeds, as confirmed by one-way ANOVA statistical tests after 15,000 observations. Furthermore, Fig. 9 illustrates that the ORS-APHS algorithm, utilizing the greedy heuristic initialization method and innovative heuristic neighborhood creation rule, converges to a superior solution more rapidly than the other three combinations, achieving this after approximately 6,000 observations. This indicates that employing both the greedy heuristic initialization method and the innovative heuristic neighborhood creation rule significantly improves the ORS-APHS algorithm’s performance, both in terms of solution quality and

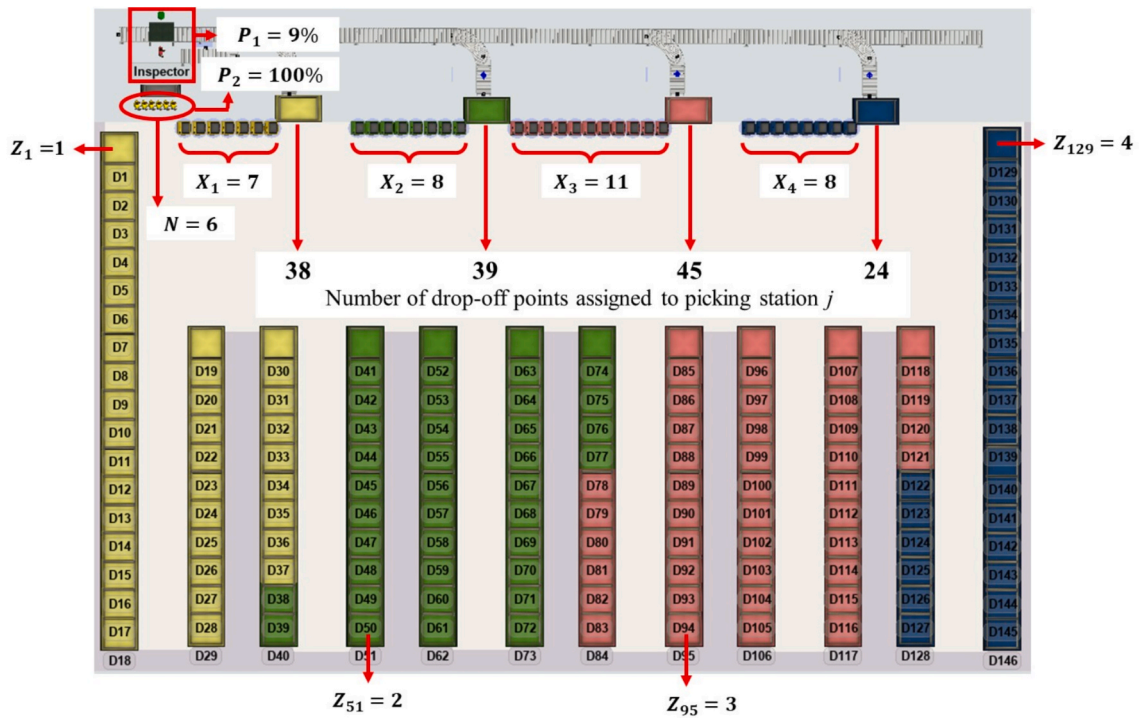


Fig. 6. The optimal tactical design of robotic sorting system with guaranteed security level for the case study.

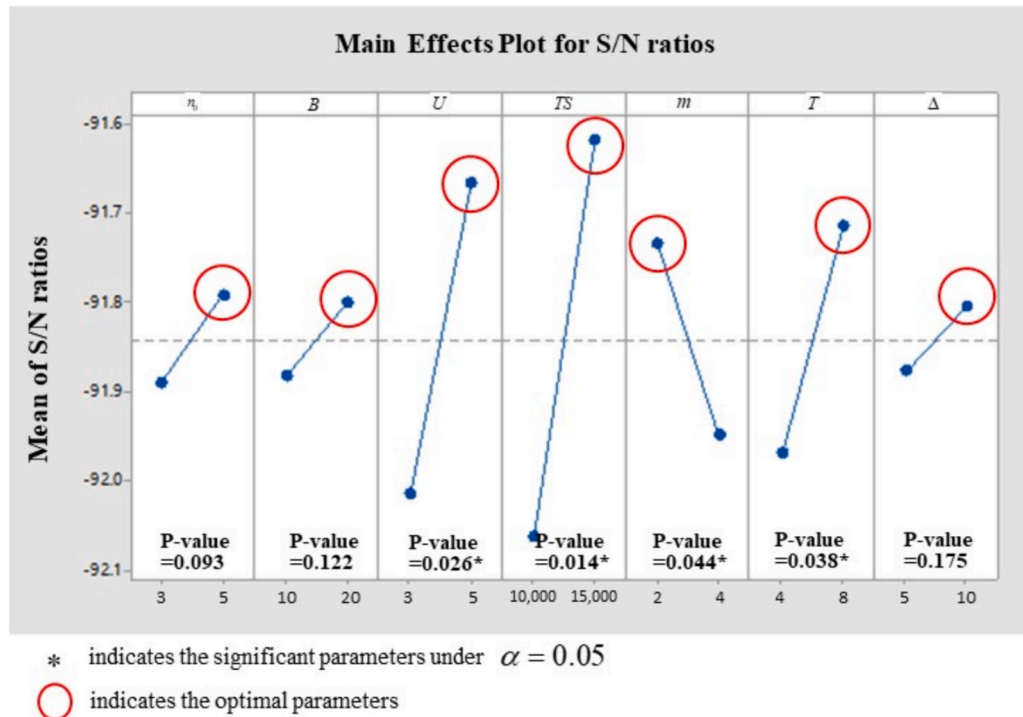


Fig. 7. Main effect plot for the parameter analysis of the ORS-APHS algorithm.

computational time. Moreover, the heuristic neighborhood creation rule has more contribution to the objective value than the greedy heuristic initialization method because ORS-APHS with only random initialization strategy outperforms ORS-APHS with only random neighborhood strategy.

5.2.3. Comparison between ORS-APHS and other existing algorithms

To assess the ORS-APHS algorithm's effectiveness, we compare its performance with established algorithms such as the Genetic Algorithm (GA) and Tabu Search (TS). Both GA and TS utilize mixed integer and continuous encoded solution representations to solve the tactical design problems of the RSS system. Integer encoding is used to determine decisions related to drop-off point assignment, AMRs allocation, and re-

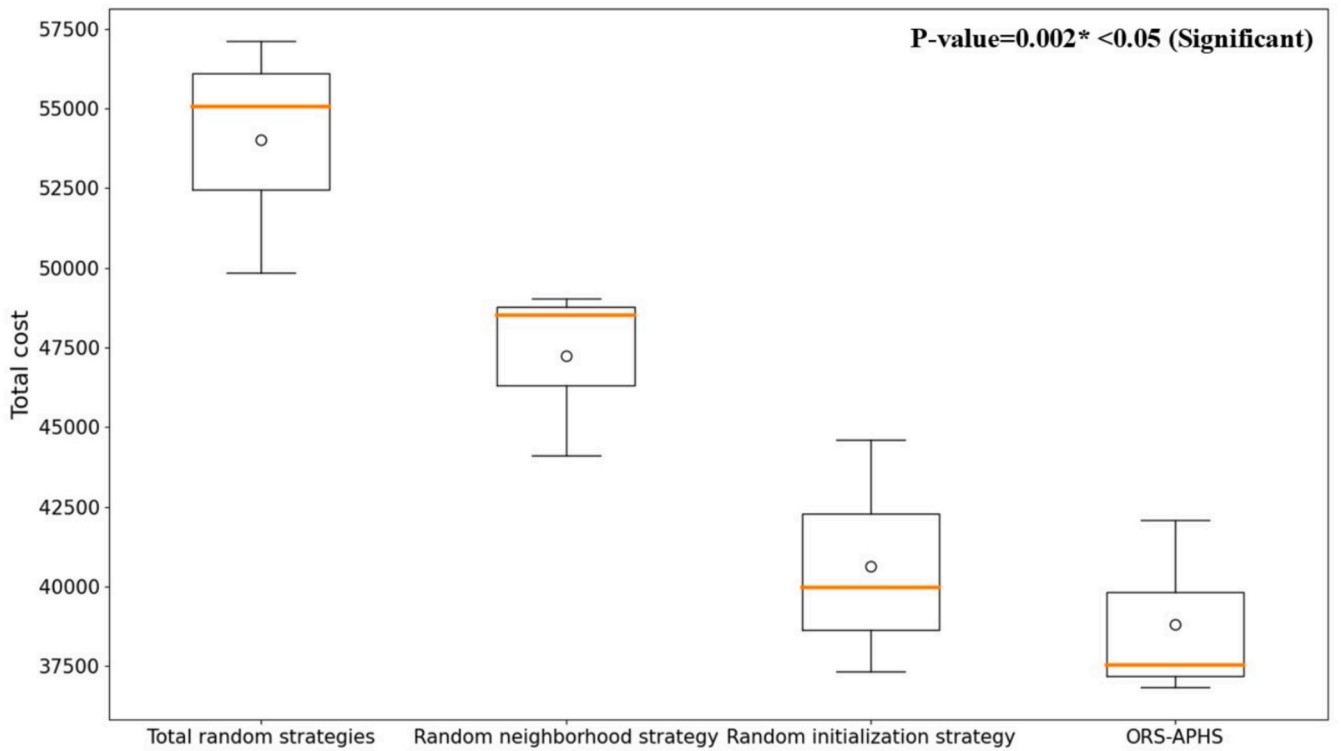


Fig. 8. Performance comparison among various initialization strategies and neighborhood rules employed by ORS-APHS through one-way ANOVA statistical tests.

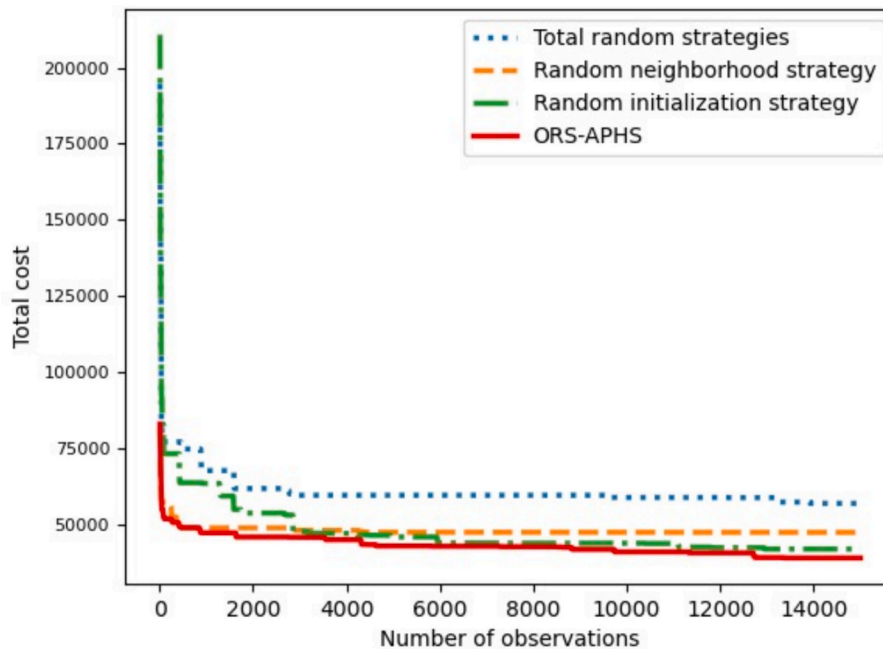


Fig. 9. Performance of various initialization and neighborhood strategies adopted by ORS-APHS.

inspection personnel allocation, while continuous encoding between 0 and 1 is used to determine the inspection probabilities at the initial screening and physical re-inspection stations. GA, a population-based search heuristic, mimics natural evolutionary processes, including “survival of the fittest”, crossover, and mutation, to identify optimal solutions in combinatorial optimization problems. To solve this tactical design problems, we adopt single-point crossover and multi-point crossover as crossover operators, along with partition randomization mutation and swap mutation as mutation operators to assist the GA

evolution process in generating new offspring. In addition, using the Taguchi method of experimental design, we determined the optimal parameters for the GA algorithm by exploring a crossover rate range of 0.6 to 0.9, a mutation rate range of 0.01 to 0.1, and population sizes of 50, 100, and 150 (Spall, 2005). The best parameters identified were a population size of 100, a crossover rate of 0.7, and a mutation rate of 0.05, which were then used for comparison with the ORS-APHS algorithm. These parameters are optimized to allow GA to both explore and exploit the solution space effectively, thereby enhancing the likelihood

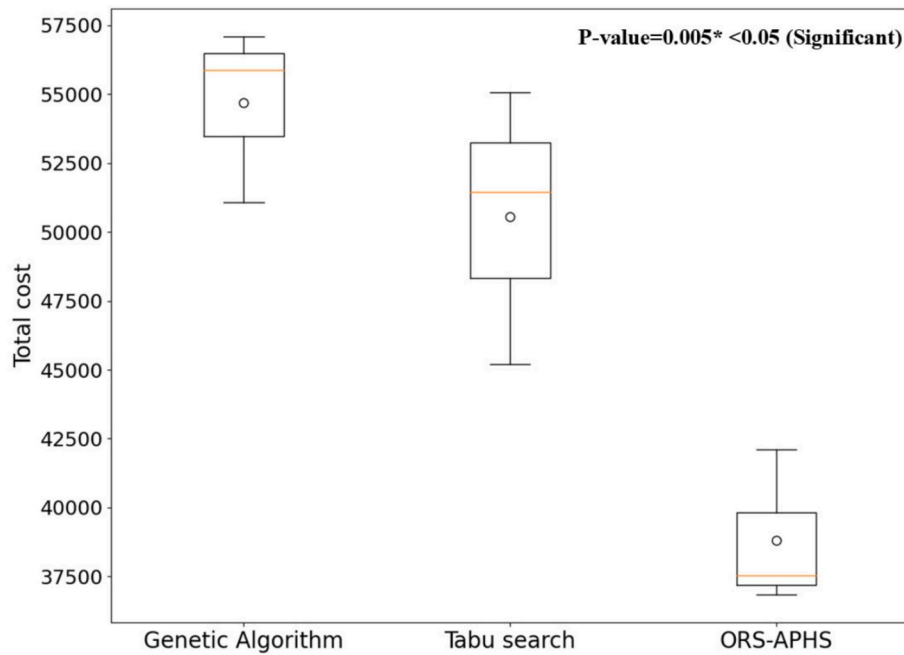


Fig. 10. Performance comparison between ORS-APHS and other existing algorithms through one-way ANOVA statistical tests.

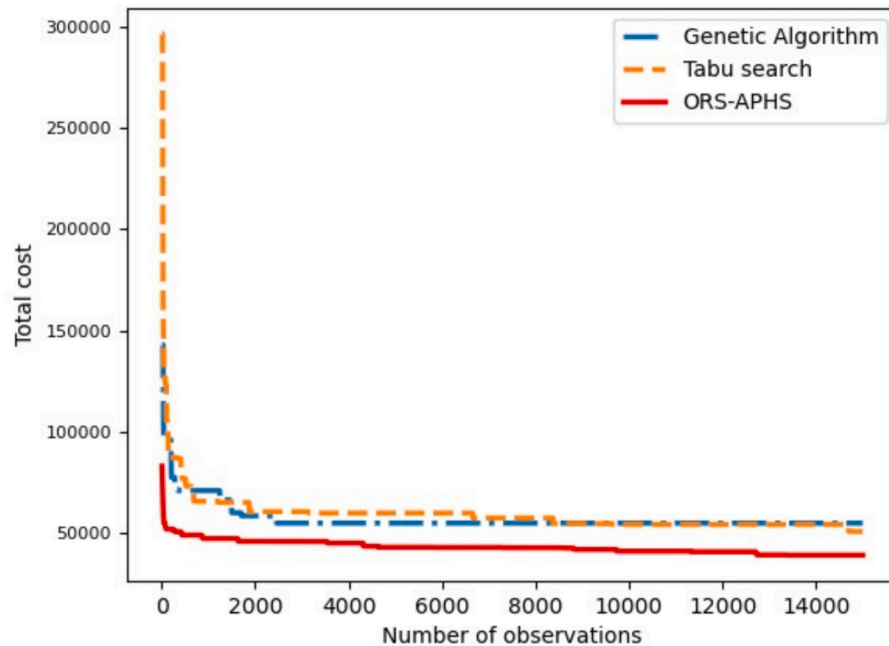


Fig. 11. Convergence analysis comparing Genetic Algorithms, Tabu Search, and ORS-APHS.

of identifying a near-optimal solution. TS operates as a single-point random search algorithm that strategically explores the solution space by permitting transitions to neighboring solutions, even if they temporarily worsen the objective value, while preventing revisits to previous solutions using a mechanism known as the tabu list. In this study, we determined the optimal tabu list size to be 20, based on the Taguchi method of experimental designs ranging from 10 to 40 (Spall, 2005). In addition, the TS algorithm adopts the heuristic neighborhood creation rules proposed in this study (see Section 4.4) to generate neighboring candidate solutions, from which the best neighboring solution is selected as the new solution after the next move. The initial solutions for both GA and TS are generated randomly. During the search for the optimal

solution, both GA and TS algorithms perform an equal 10 simulation replications for each solution to estimate its objective value, serving as the basis for evaluating solution quality.

Due to the stochastic nature of the algorithms, we conducted 30 repeated experiments with different random seeds for each of the three algorithms to calculate and estimate their average objective values, which are visually represented in the box plots shown in Fig. 10. As can be seen in Fig. 10, statistical analysis via one-way ANOVA with $P\text{-value} = 0.005 \leq 0.05$ after 15,000 observations confirms that the ORS-APHS algorithm significantly outperforms the other algorithms, GA and TS, in terms of average total cost and variance across 30 repeated experiments with different random seeds. The convergence analysis

depicted in Fig. 11 illustrates that the ORS-APHS algorithm rapidly achieves a lower objective value after 2,000 simulation observations. Consequently, ORS-APHS not only outperforms both GA and TS, but it also requires fewer simulation observations and budgets to reach near-optimal solutions. Particularly noteworthy is that while GA is typically better suited for discrete problems, ORS-APHS efficiently uses OCBA- m to allocate simulation replications, thus reducing overall simulation costs (replications), and enhances its ability to find near-optimal solutions through the APHS component with the mechanisms of global exploration and local exploitation. Furthermore, as shown in Fig. 10, the TS algorithm outperforms the GA algorithm due to its adoption of the innovative neighborhood creation rules proposed in this study for identifying high-quality neighboring solutions. This highlights the significant contribution of the heuristic neighborhood creation rules developed in this research to improving the performance of the TS algorithm.

5.2.4. Discussion

Based on the experimental results from the previous sections, it can be summarized that ORS-APHS has three major advantages that make it outperform other algorithms: (1) generating high-quality initial solutions based on the problem structure, (2) combining global exploration and local exploitation to produce promising and high-performing neighboring solutions, and (3) efficiently allocating simulation costs (replications) to evaluate and screen for high-quality feasible solutions, thereby avoiding the waste of simulation resources on inferior solutions. Firstly, as shown in Figs. 9 and 11, the greedy heuristic initialization method in ORS-APHS is capable of generating initial solutions with significantly higher quality compared to other algorithms (such as GA or TS) during the initial phase of the algorithm. This is primarily because the greedy heuristic initialization method is specifically designed based on the characteristics of the RSS system's tactical design problem. As a result, ORS-APHS inherently has a performance advantage, significantly outperforming other algorithms. Secondly, as observed from Figs. 9 and 11, the total cost objective value of the ORS-APHS algorithm demonstrates a rapid downward trend during the initial simulation observations. This indicates that starting from high-quality initial solutions, the ORS-APHS algorithm can quickly identify highly advantageous optimal feasible solutions within the first few simulation observations. This is primarily attributed to the innovative neighborhood creation rules proposed in this study, which generate high-quality neighboring integer variable solutions (N, X, Z), and the APHS mechanism, which employs global exploration and local exploitation to search for optimal

neighboring solutions for continuous variables (P_1, P_2) in the most promising regions. Similarly, since the TS algorithm incorporates the innovative neighborhood creation rules proposed in this study, its performance surpasses that of the GA algorithm. Finally, the ORS-APHS algorithm adopts an OCBA- m -based screening mechanism, effectively allocating simulation replications to promising and high-quality solutions for evaluating objective function values and selecting the top m optimal solutions to facilitate subsequent searches. This mechanism avoids wasting simulation resources on evaluating inferior solutions, enabling OCBA- m to save more simulation replications and resources, providing greater opportunities for in-depth exploration of near-optimal solutions. In contrast, the GA and TS algorithms evaluate the objective values of each solution using a fixed and equal number of simulation replications, regardless of the solution's quality. This leads to simulation resources being wasted on evaluating inferior solutions, leaving no additional resources for deeper exploration. This is one of the reasons why ORS-APHS outperforms GA and TS in terms of performance. In addition, ORS-APHS provides a generalized solution framework for mixed-integer simulation optimization problems. This framework can be applied not only to design decision problems for RSS systems with other layouts (such as grid-type or football field configurations) but also to other stochastic optimization problems involving both integer and continuous variables. By redesigning the initial solution generation method and the neighborhood solution creation rules based on the specific characteristics of the problem and integrating them into the ORS-APHS framework, the efficiency and effectiveness of the solution process can be further enhanced.

5.3. Sensitivity analysis

In this section, we perform a sensitivity analysis on the impact of changes in system security levels, variations in the number of available AMRs, adjustments in physical re-inspection times, and distributions of different drop-off points for arriving parcels on the objective values. All experiments are conducted based on the case study of a robotic sorting system for international inbound parcel processing, as described in Section 5.1. For each scenario, we perform 30 simulation replications to assess the optimal solutions of the tactical design decisions derived from the ORS-APHS, aiming to calculate their expected objective value (total expected cost) and illustrate the results using box plots for detailed analysis. Subsequently, statistical tests are used to examine whether significant differences exist among the various parameter levels.

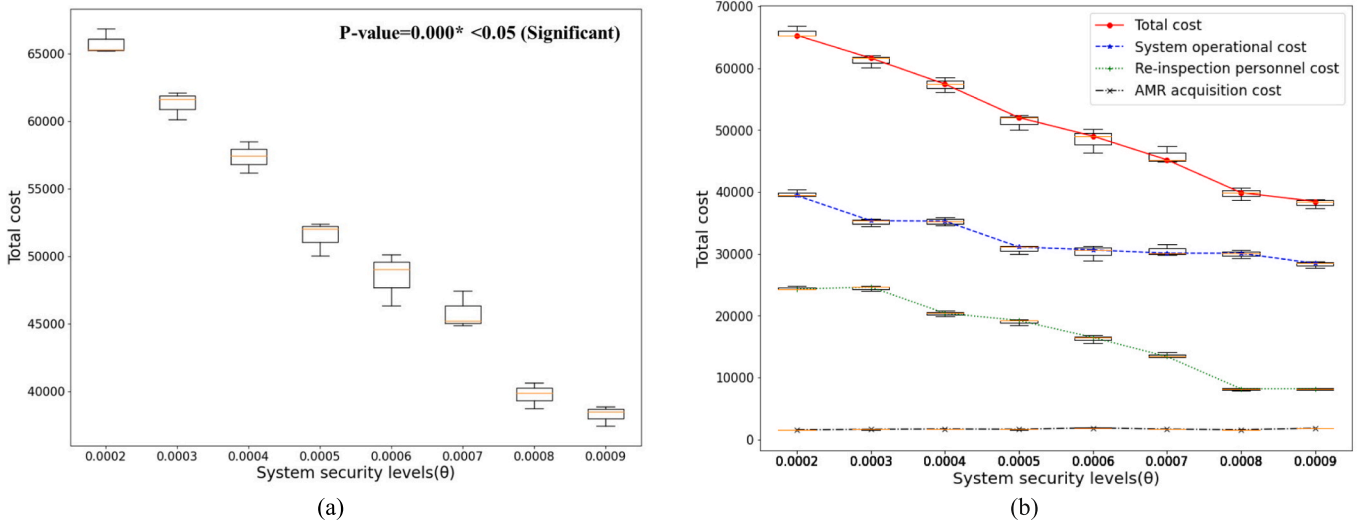


Fig. 12. Comparison of total costs and various cost elements among different levels of system security levels through one-way ANOVA statistical tests.

Table 5

The inspection probability settings at the initial screening and physical re-inspection stations under various system security levels.

System security level (θ)	0.0002	0.0003	0.0004	0.0005	0.0006	0.0007	0.0008	0.0009
P_1	0.73	0.64	0.55	0.45	0.38	0.34	0.18	0.09
P_2	1	1	0.99	1	0.99	0.94	1	1
$P_1 \times P_2$	0.73	0.64	0.5445	0.45	0.3762	0.3196	0.18	0.09

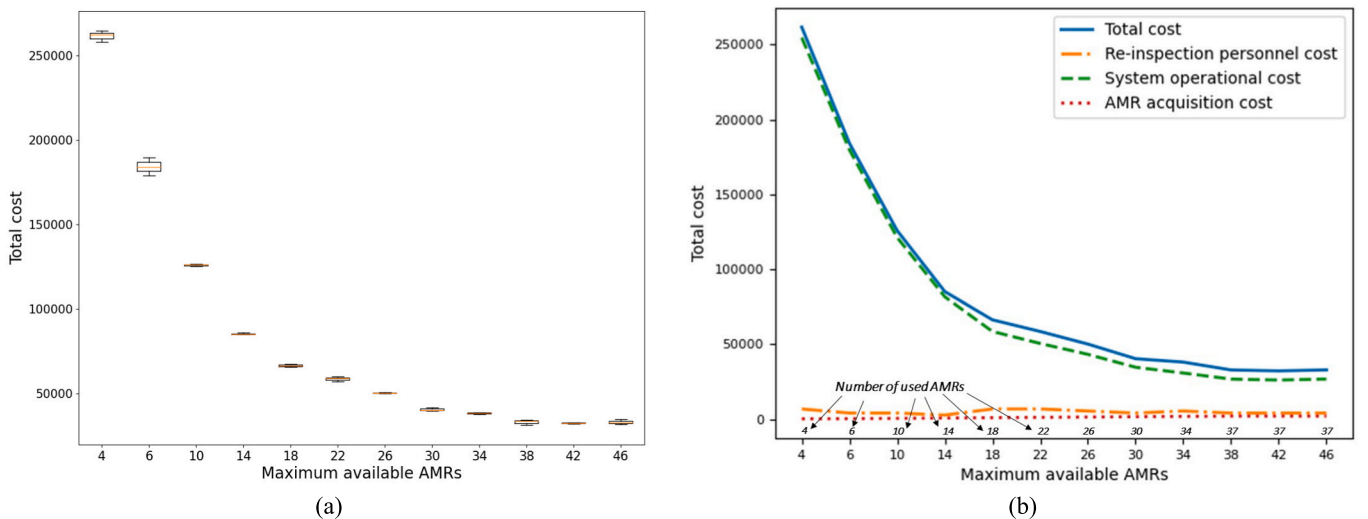
5.3.1. Influence of different system security levels

To successfully prevent the inflow of illicit goods through international postal parcel operations and mitigate potential national security and health risks, the system security level is a critical parameter that affects the efficiency and safety of the robotic sorting system with guaranteed security level. Therefore, we investigate the effects of setting different system security level thresholds, ranging from a low threshold setting of 0.0002 (indicating a higher system security level) to a high threshold setting of 0.0009 (indicating a lower system security level), on the total costs and various cost elements. Fig. 12(a) illustrates that as the system security threshold increases (signifying reduced security), the total cost required for the RSS system shows a downward trend. The one-way ANOVA statistical tests with P -value = $0.000 \leq 0.05$ also reveal significant differences in total cost objectives across various system security thresholds. As a result, it also suggests that placing greater emphasis on system security to block the influx of illegal parcels (by lowering the security threshold) results in increased costs for implementing the tactical design of the RSS system with guaranteed security level. As shown in Fig. 12(b), the rise in total costs is primarily due to increased re-inspection personnel expenses and higher system operational costs. This cost increment is attributed to the need for more personnel at the physical re-inspection stations and extended durations for parcel sorting and inspection in the RSS systems designed with higher security levels. Moreover, the results for the inspection probability settings at the initial screening and physical re-inspection stations, as shown in Table 5, indicate that the re-inspection personnel at the physical re-inspection station must consistently maintain a stringent approach (with P_2 exceeding 95 %), regardless of the system security level threshold. On the contrary, the setting of the system security level threshold mainly influences the initial inspection probability P_1 at the initial screening station. For enhanced system security (indicated by lower security threshold settings), it is necessary to increase the initial

screening inspection probabilities (such as purchasing higher-grade X-ray machines [58], improving the image interpretation capabilities of X-ray operators, or adopting AI-enabled X-ray image detection technologies [12] etc.), ensuring rigorous inspections are conducted early on to minimize the likelihood of illegal parcels moving forward in the AMR sorting process. Based on the results shown in Fig. 12, the tactical design decision-makers of the RSS system can know how to select appropriate system security level thresholds to determine simultaneously drop-off point assignment, AMRs allocation, re-inspection personnel allocation, and inspection probability setting decisions to minimize total costs and maximize the sorting efficiency of the RSS system. Setting the system security threshold too low can significantly increase the total cost, while setting it too high can make the RSS system very insecure.

5.3.2. Influence of different maximum available AMRs

In the international inbound parcel sorting operations enabled by the robotic sorting system, AMRs are vital and costly resources. A shortage of AMRs greatly diminishes the sorting efficiency of the RSS system, leading to extended parcel sorting times. This affects the established outbound truck schedule at the postal logistics center, resulting in increased overtime costs for trucks [20]. To understand the impact, we examine the effect of changing the number of available AMRs, ranging from a minimum of 4 AMRs to a maximum of 46 AMRs. Fig. 13 (a) illustrates that as the maximum number of available AMRs increases, there is a corresponding decline in the total cost for the RSS system. Fig. 13(b) shows that the primary factor leading to the reduction in total costs is the significant decrease in system operational time and system operational costs due to the increased number of AMRs. The one-way ANOVA statistical tests with P -value = $0.000 \leq 0.05$ also demonstrates significant differences in the total cost targets across various levels of maximum available AMRs. It is noteworthy that when the number of available AMRs is low, increasing their quantity significantly

**Fig. 13.** Comparison of total costs and various cost elements under different levels of maximum available AMRs through one-way ANOVA statistical tests.

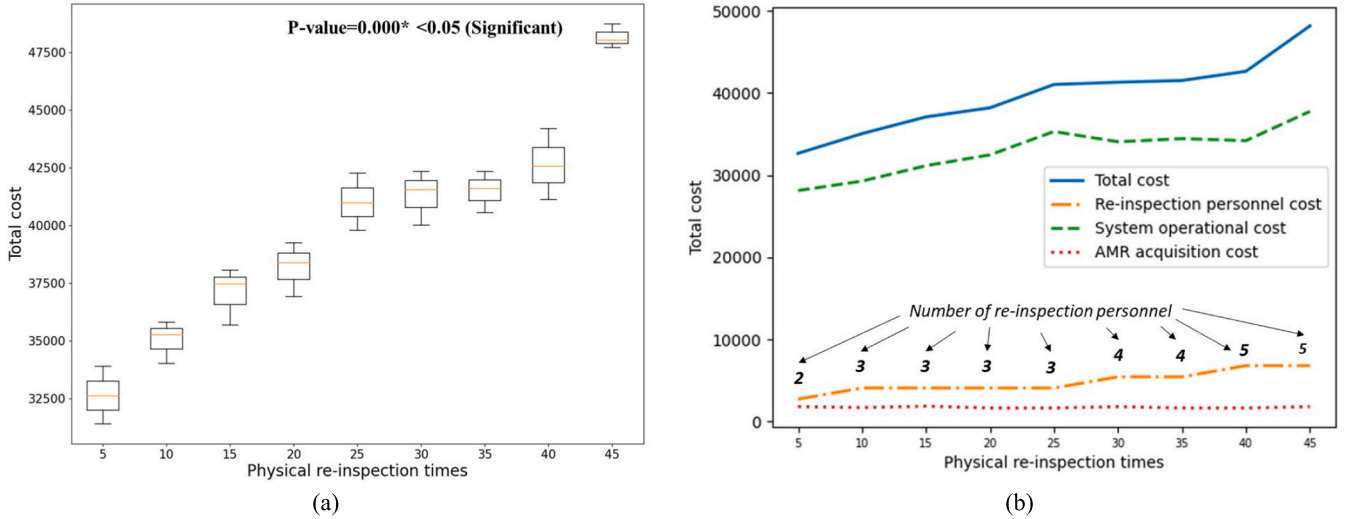


Fig. 14. Comparison of total costs and various cost elements under different levels of physical re-inspection times through one-way ANOVA statistical tests.

reduces total costs. However, when the number of available AMRs is already high, additional increases only lead to a marginal decrease in total costs. Thus, an excessive number of AMRs does not substantially improve system efficiency or operational time; in fact, too many AMRs not only increase procurement costs but also the potential for system congestion, thereby extending operational times. Nevertheless, even with an excessive number of maximum available AMRs, the proposed ORS-APHS algorithm can still identify the optimal number of AMRs actually required to minimize system operational time and costs. As shown in Fig. 13(b), when the number of maximum available AMRs exceeds 38, the optimal number of AMRs for the system remains at 37, indicating that the additional AMRs are unnecessary and would reduce system efficiency. Based on the results shown in Fig. 13, the tactical design decision-makers of the RSS system can determine the optimal number of used AMRs under the given maximum available AMRs to achieve target performance and meet the established outbound truck schedule [20], while reducing waste caused by excessive AMRs.

5.3.3. Influence of different physical re-inspection times

In addition to the availability of AMRs, the physical re-inspection time is another parameter that significantly affects the sorting efficiency of the RSS system and the required number of re-inspection

personnel. Therefore, we explore the impact of implementing different physical re-inspection times for each re-inspected parcel, ranging from a minimum of 4 s to a maximum of 45 s, on total costs and the number of re-inspection personnel. As shown in Fig. 14(a), the total cost of the RSS system increases with increasing physical re-inspection times. The one-way ANOVA statistical tests with $P\text{-value} = 0.000 \leq 0.05$ indicate that the objective values under different physical re-inspection times have significant differences. Fig. 14(b) shows that the increase in total costs is due to a rise in system operational costs and higher expenses for re-inspection personnel (from 2 to 5 people). This phenomenon indicates that longer inspection times by re-inspection personnel for suspected illegal parcels, although potentially indicating a more thorough approach, lead to prolonged system operation times, reducing sorting efficiency and requiring additional personnel for inspection tasks. Consequently, how to train re-inspection personnel to detect illicit parcels in shorter times while maintaining rigorous standards is an important decision-making issue in this RSS system.

5.3.4. Influence of different drop-off point distributions for arriving parcels

The destinations of incoming international parcels vary, resulting in distinct sorting requirements at each drop-off point, termed as drop-off point distributions for arriving parcels, which generally adhere to a

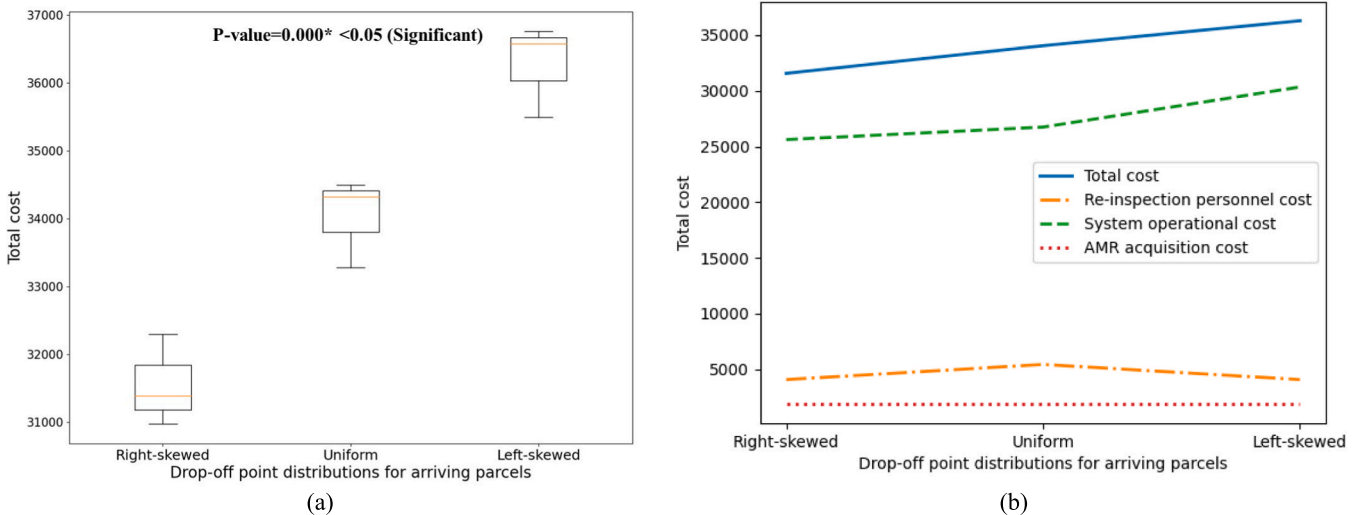


Fig. 15. Comparison of total costs and various cost elements under various drop-off point distributions for arriving parcels through one-way ANOVA statistical tests.

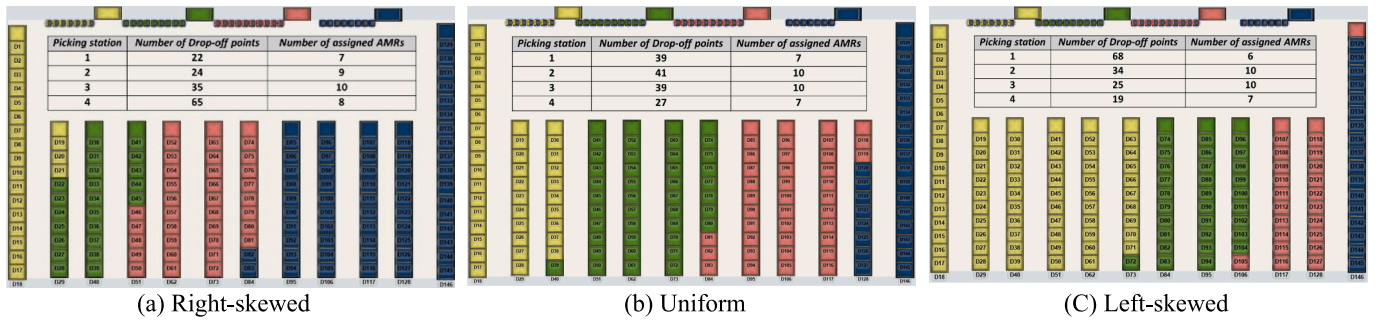


Fig. 16. Optimal drop-off point assignment solutions under various drop-off point distributions for arriving parcels.

Multinoulli distribution depicted in Fig. A3 in Section A3 of Appendix A in the case study. Therefore, this section investigates how various drop-off point distribution patterns influence the RSS system's total costs and tactical design decisions. We analyze three distinct drop-off point distribution types: right-skewed, uniform, and left-skewed distributions. In a right-skewed distribution, the demand for sorting at each drop-off point diminishes as their numeric order increases, with the points nearer to the layout's left side experiencing higher demands. In contrast, a left-skewed distribution sees an increase in sorting demand as the drop-off point number increases, placing higher demands on points located towards the right side of the layout. Conversely, a uniform distribution maintains consistent sorting demands across all drop-off points. According to Fig. 15(a), total costs are minimized under a right-skewed distribution and maximized under a left-skewed distribution. Fig. 15(b) explains that the left-skewed distribution incurs higher total costs as it necessitates longer conveyance of parcels to more remote picking stations, thereby elevating system operational times and costs. Fig. 16 illustrates the optimal drop-off point assignment solutions Z under different drop-off point distributions for arriving parcels, derived from the proposed ORS-APHS algorithm. The results demonstrate that the ORS-APHS algorithm efficiently assigns fewer drop-off points with higher sorting demands located on the left side of the layout to picking station 1 in a right-skewed distribution scenario. In contrast, it allocates more drop-off points with lower sorting demands on the right to picking station 4, optimizing for the lowest total costs. Conversely, in a left-skewed distribution scenario, the optimal drop-off point assignments are the opposite, reflecting the distribution's characteristics. Consequently, these findings offer critical guidance for RSS system tactical design decision-makers on how to allocate parcel destinations to drop-off points effectively [28]. For substantial reductions in system operational times and costs, it is beneficial to assign destinations with higher sorting demands to drop-off points situated on the left side of the layout.

5.3.5. Discussion

Based on the results of the sensitivity analysis, we found that the setting of system security thresholds is a critical factor affecting the operational costs of the RSS system with guaranteed security level and its ability to combat illegal trade (e.g., inspection probability settings at the initial screening and physical re-inspection stations). In contrast, other factors (such as the maximum number of available AMRs, physical re-inspection time, and drop-off point distribution for arriving parcels) primarily influence the system's operational costs. This finding highlights to decision-makers that designing a more secure RSS system requires additional investment in hardware and training at the initial screening and physical re-inspection stations to enhance the rigor of

inspections. However, increasing inspection rigor may extend the time required for thorough inspections by re-inspection personnel (i.e., physical re-inspection time), necessitating an increase in personnel to maintain operational efficiency. To effectively combat illegal trade, another critical supporting measure is to implement comprehensive training and certification programs for the re-inspection personnel. These programs should include familiarization with inspection processes and rigorous testing to enhance efficiency (reducing physical re-inspection time) and the ability to detect illegal goods. This conclusion aligns with discussions held with the case company and customs authorities and is consistent with their future plans and initiatives.

Furthermore, from the sensitivity analysis presented in earlier sections, we also summarize the following managerial insights and guidelines to support decision-makers in making more informed tactical design decisions for the RSS system. It is important to note that these conclusions are specifically applicable to the "E-shaped" structure layout used in the Chunghwa Post case study. In the future, the same analytical methods can be applied to other RSS layouts, such as grid-type or football field configurations.

1. Setting appropriate system security thresholds

The system security threshold is a critical factor influencing tactical design decisions for the RSS system. It not only determines the overall operational costs but also directly affects the inspection probabilities at the initial screening and physical re-inspection stations, reflecting the level of inspection rigor. Lower security thresholds (i.e., stricter inspections) are more effective in combating illegal goods or trade but significantly increase operational costs. Conversely, higher thresholds may reduce costs but compromise the security of the RSS system, potentially allowing illegal goods to pass through. Tactical design decision-makers can utilize the sensitivity analysis results to select an appropriate system security threshold that balances security and efficiency in the RSS system.

2. Determining the optimal number of AMRs to purchase

Identifying the optimal number of AMRs is crucial. Too few AMRs significantly reduce the operational efficiency of the RSS system and increase operational costs. On the other hand, purchasing too many AMRs not only raises procurement costs but may also lead to potential system congestion, resulting in efficiency losses. Additionally, decision-makers might face unnecessary idle costs from unused AMRs purchased to avoid congestion.

3. Optimizing re-inspection personnel training programs

Revising the training programs for re-inspection personnel is essential to enable them to maintain 100 % strict standards while detecting illegal parcels in less time. This approach minimizes labor costs while ensuring thorough inspections, ultimately reducing the operational costs of the system.

4. Dynamically optimal adjustment of the drop-off point assignment solutions

Since the drop-off point distributions for arriving parcels change over time, fixed drop-off point assignment solutions can lead to increased operational costs. To address this, it is necessary to periodically optimize and adjust the drop-off point assignments using the ORS-APHS algorithm proposed in this study. Ideally, destinations with higher sorting demands should be assigned to drop-off points located on the left side of the layout, which can reduce operational costs and improve overall efficiency.

6. Conclusions

The surge in online e-commerce transactions and the promotion of free trade have significantly increased the flow of international parcels, easily surpassing the capacity of existing governance mechanisms and creating numerous opportunities for illicit trade. This illicit trade activity has led to various illegal goods, including counterfeit products, drugs, illegal narcotics, wildlife, human trafficking, and illegal weapons, easily entering different countries through cross-border transportation or their illicit supply chain systems, causing adverse impacts on the social economy, national security, public health, the environment, and politics. Most governments have primarily focused on preventing illicit trade by disrupting illegal supply chain networks (e.g., node disruption, link disruption, etc.) through proactive law enforcement crackdowns and the formulation of laws and policies to raise consumer awareness. However, they have largely neglected the use of emerging logistics and inspection technologies to improve or design secure logistics systems that can effectively detect and prevent the entry of illicit international goods, which is especially crucial for postal service systems with outdated parcel handling processes and equipment. Therefore, in this study, we address the tactical design decisions for a robotic sorting system with a guaranteed security level, integrating multi-stage security screening into the parcel sorting process to effectively combat the influx of illicit goods or parcels through international import operations. A mixed-integer stochastic optimization model is proposed to derive the optimal tactical design solutions, including the number of required AMRs, the number of inspection personnel, the allocation of drop-off points at each picking station, and the security screening probabilities at each inspection stage, aiming to minimize overall system costs while meeting a specified security threshold. Then, due to the analytical intractability of the proposed model's objective function, we develop a novel and efficient mixed-integer simulation optimization algorithm, namely the ORS-APHS, which combines a high-resolution RSS stochastic simulation model to tackle the mixed-integer and continuous variable tactical design decisions for the robotic sorting system with guaranteed security levels under a limited simulation budget.

To illustrate the validity and usefulness of our developed model and algorithm, we cooperated with Chunghwa Post company in Taiwan to conduct an empirical study using the design of the international import parcel sorting system in their new mail processing center. The empirical

results demonstrate that the ORS-APHS algorithm has superior convergence performance compared to two commonly used existing algorithms, Genetic Algorithm and Tabu Search, under the same simulation budget. Moreover, simultaneously employing the greedy heuristic initialization method and the innovative heuristic neighborhood creation rule proposed in this study significantly improves the ORS-APHS algorithm's performance, both in terms of solution quality and computational time, compared to the random initialization method and random neighborhood solution generation method. Specifically, the innovative heuristic neighborhood creation rule contributes more to the objective value than the greedy heuristic initialization method. Through sensitivity analysis, we found that different changes in the predetermined system security levels, variations in the number of available AMRs, adjustments in physical re-inspection times, and distributions of different drop-off points for arriving parcels all have a significant influence on total system costs and tactical design solutions. Among them, the predetermined system security level settings have a very significant impact on performance metrics such as re-inspection personnel costs, system operational costs, and total system costs, as well as the inspection rigor (i.e., inspection probability settings) at different inspection stations. The predetermined system security level settings have a trade-off relationship with the total system cost. To enhance system security (indicated by lower security threshold settings), the RSS system must increase the inspection probability at each checkpoint, which will necessitate more re-inspection personnel and extend the parcel sorting operation time, thereby increasing re-inspection personnel costs, system operational costs, and total system costs. Conversely, setting the system security threshold too high can significantly reduce total costs and related inspection costs but will make the RSS system very insecure. Consequently, tactical design decision-makers can use our sensitivity analysis results to select an appropriate system security level threshold to achieve both security and efficiency in the RSS system. Furthermore, autonomous mobile robots are the critical resource with the greatest impact on the system operation time and total cost of the RSS system. When the number of maximum available AMRs is low, the system operational time and total costs will significantly increase, but when the number is too high, although these costs decrease, an excessive number of AMRs may increase the possibility of system congestion, thereby extending operational time. Nonetheless, in scenarios where the number of maximum available AMRs is excessive, the ORS-APHS algorithm developed in this study can accurately determine the optimal number of AMRs needed, offering decision-makers valuable insights to minimize system operational time and costs. Finally, we found that the patterns of drop-off point distributions for arriving parcels also impact total system costs, thus providing critical guidance for RSS system tactical design decision-makers on effectively allocating parcel destinations to drop-off points in the future [28]. The managerial insights and guidelines obtained from sensitivity analysis are highly valuable and constructive for the decision-makers in designing the robotic sorting system with guaranteed security level. These managerial insights include: (1) setting appropriate thresholds of system security level, (2) determining the optimal number of AMRs to purchase, (3) optimizing the training programs for re-inspection personnel, and (4) developing the best drop-off point allocation adjustment plan based on different parcel drop-off point distributions, to design an RSS system that is both secure and cost-efficient.

There are four topics for future exploration. First, we can relax the two main assumptions of this study for further investigation. The first assumption is that each AMR is fixed to serve a specific picking station (dedicated strategy). Future research can explore the impact of adopting

other strategies (e.g., Random, Closest, or Shortest-robot-queue) or develop optimal dynamic allocation strategies [75]. The second assumption is that each drop-off point handles only one destination, while in reality, destinations often exceed drop-off points in an RSS system. Thus, decisions on which destinations should be assigned to which drop-off points or how many destinations should be assigned to each drop-off point can be explored. Second, while this study focuses solely on the tactical design decisions for the robotic sorting system with guaranteed security level, future research should explore the integration of these tactical-level decisions with real-time operational-level decisions [56,28,35,69], such as AMR/robot scheduling and path planning, to achieve synchronized planning and enhance the overall safety and efficiency of the RSS system. Third, due to the high computational complexity and analytical intractability of the objective function in the tactical design problem of the RSS system, the development of other more efficient mixed-integer simulation optimization algorithms is very

worth of further investigation. Fourth, we can apply the proposed ORS-APHS algorithm to solve other mixed-integer simulation optimization problems [60] to demonstrate its capabilities and benefits.

CRediT authorship contribution statement

Tzu-Li Chen: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Data curation, Conceptualization. **Yu-Xuan Li:** Validation, Software, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

A1. Proof of Proposition 1.

We first prove $\mathbf{P}\{T|A\}$ is decreasing in P_1 . Considering ρ_1 and ρ_2 as the actual specific values of the decision variable P_1 , where $\rho_1 < \rho_2$, and $f(\rho_1)$ and $f(\rho_2)$ representing the function $\mathbf{P}\{T|A\}$ at ρ_1 and ρ_2 respectively, we can then calculate:

$$\begin{aligned}
 & f(\rho_1) - f(\rho_2) \\
 &= \frac{\tau - \rho_1 \times \beta \times P_2}{1 - \rho_1 \times \beta \times P_2} - \frac{\tau - \rho_2 \times \beta \times P_2}{1 - \rho_2 \times \beta \times P_2} \\
 &= \frac{(\tau - \rho_1 \times \beta \times P_2) \times (1 - \rho_2 \times \beta \times P_2) - (\tau - \rho_2 \times \beta \times P_2) \times (1 - \rho_1 \times \beta \times P_2)}{(1 - \rho_1 \times \beta \times P_2) \times (1 - \rho_2 \times \beta \times P_2)} \\
 &= \frac{\tau - \rho_1 \times \beta \times P_2 - \tau \times \rho_2 \times \beta \times P_2 + \rho_1 \times \rho_2 \times \beta^2 \times P_2^2 - (\tau - \rho_2 \times \beta \times P_2 - \tau \times \rho_1 \times \beta \times P_2 + \rho_1 \times \rho_2 \times \beta^2 \times P_2^2)}{(1 - \rho_1 \times \beta \times P_2) \times (1 - \rho_2 \times \beta \times P_2)} \\
 &= \frac{-\rho_1 \times \beta \times P_2 - \tau \times \rho_2 \times \beta \times P_2 + \rho_2 \times \beta \times P_2 + \tau \times \rho_1 \times \beta \times P_2}{(1 - \rho_1 \times \beta \times P_2) \times (1 - \rho_2 \times \beta \times P_2)} \\
 &= \frac{-\beta \times P_2 \times (\rho_1 - \rho_2) + \tau \times \beta \times P_2 \times (\rho_1 - \rho_2)}{(1 - \rho_1 \times \beta \times P_2) \times (1 - \rho_2 \times \beta \times P_2)} \\
 &= \frac{(\rho_1 - \rho_2) \times \beta \times P_2 \times (\tau - 1)}{(1 - \rho_1 \times \beta \times P_2) \times (1 - \rho_2 \times \beta \times P_2)}
 \end{aligned}$$

First, we analyze the numerator in Eq. (EC.1). Since β and P_2 are positive while $(\rho_1 - \rho_2)$ and $(\tau - 1)$ are negative, the numerator, $(\rho_1 - \rho_2) \times \beta \times P_2 \times (\tau - 1)$, in Eq. (EC.1) is positive. Next, analyzing the denominator in Eq. (EC.1), since $\rho_1 \times \beta \times P_2$ and $\rho_2 \times \beta \times P_2$ are both less than 1, $(1 - \rho_1 \times \beta \times P_2)$ and $(1 - \rho_2 \times \beta \times P_2)$ are both positive, making the denominator, $(1 - \rho_1 \times \beta \times P_2) \times (1 - \rho_2 \times \beta \times P_2)$, in Eq. (EC.1) positive as well. Combining this with the fact that both the numerator and the denominator in Eq. (EC.1) are positive, it follows that $f(\rho_1) - f(\rho_2)$ is positive ($f(\rho_1) - f(\rho_2) > 0$). Therefore, $f(\rho_1) > f(\rho_2)$ results in $\mathbf{P}\{T|A\}$ decreasing with P_1 . Similarly, it can be proven that $\mathbf{P}\{T|A\}$ is decreasing in P_2 . Consequently, $\mathbf{P}\{T|A\}$ is a decreasing function with respect to P_1 and P_2 . This completes the proof.

A2. APHS algorithm.

Fig. A1 shows the flowchart of the APHS algorithm.

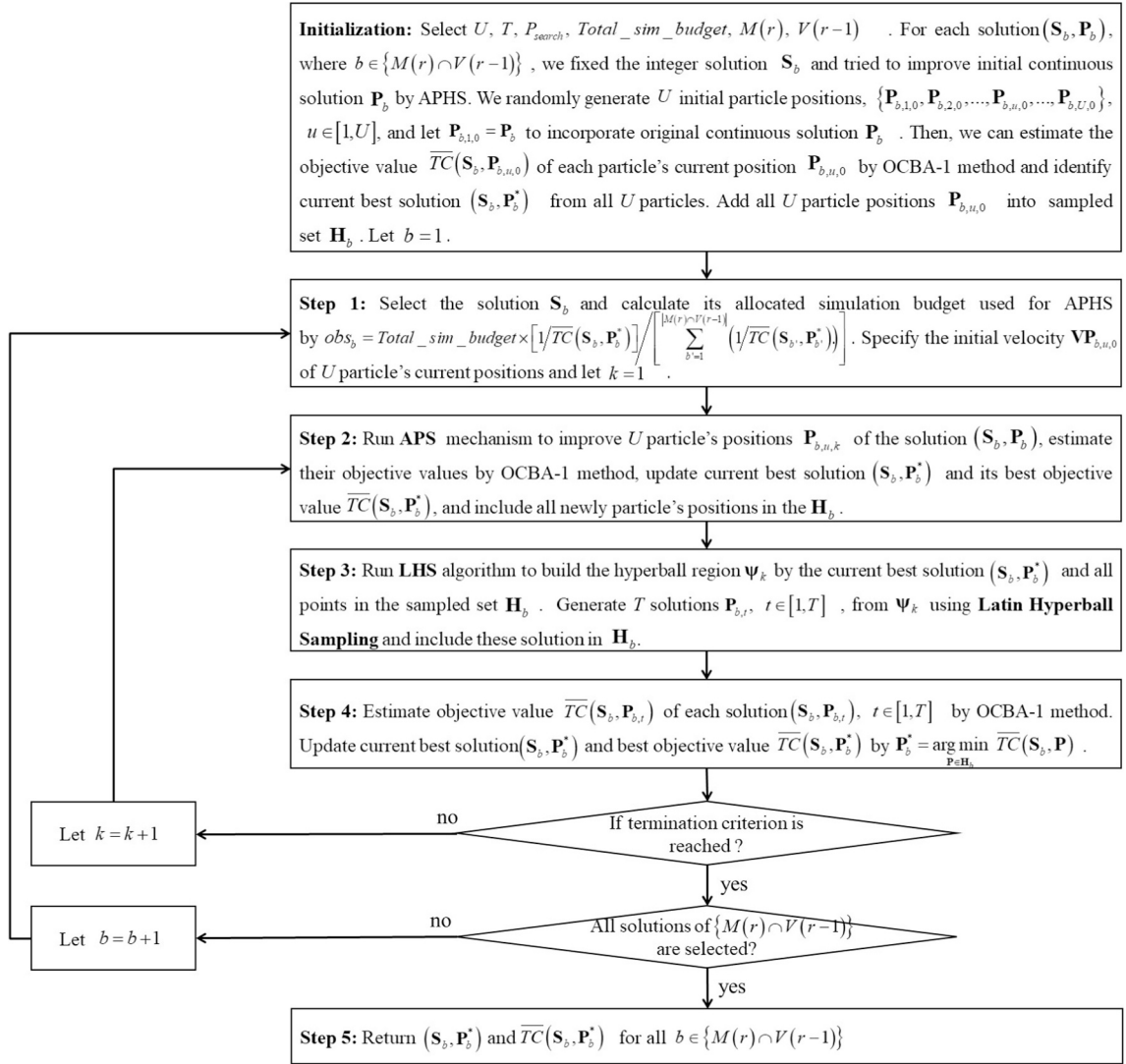


Fig. A1. Flow chart of the APHS algorithm

A3. APS algorithm.

Fig. A2 shows the flowchart of the APS algorithm.

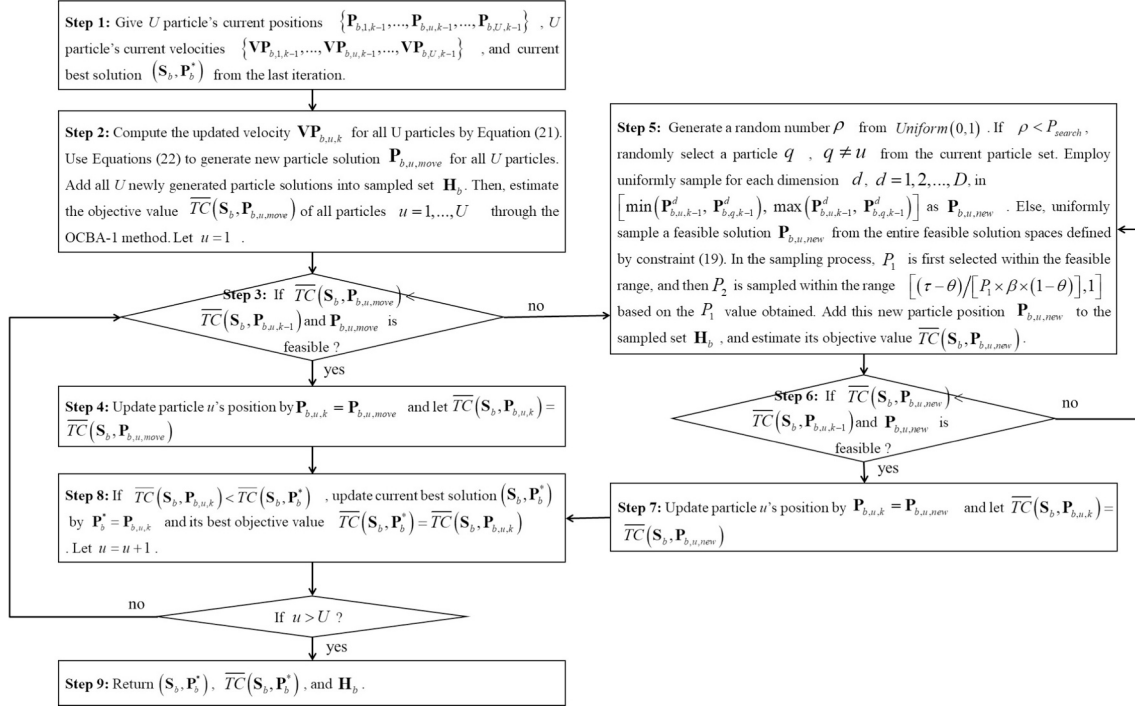


Fig. A2. Flow chart of the APS algorithm

A4. Drop-off point distribution for arriving parcels in case study.

Fig. A3 shows the multinoulli distribution of drop-off point distributions for arriving parcels.

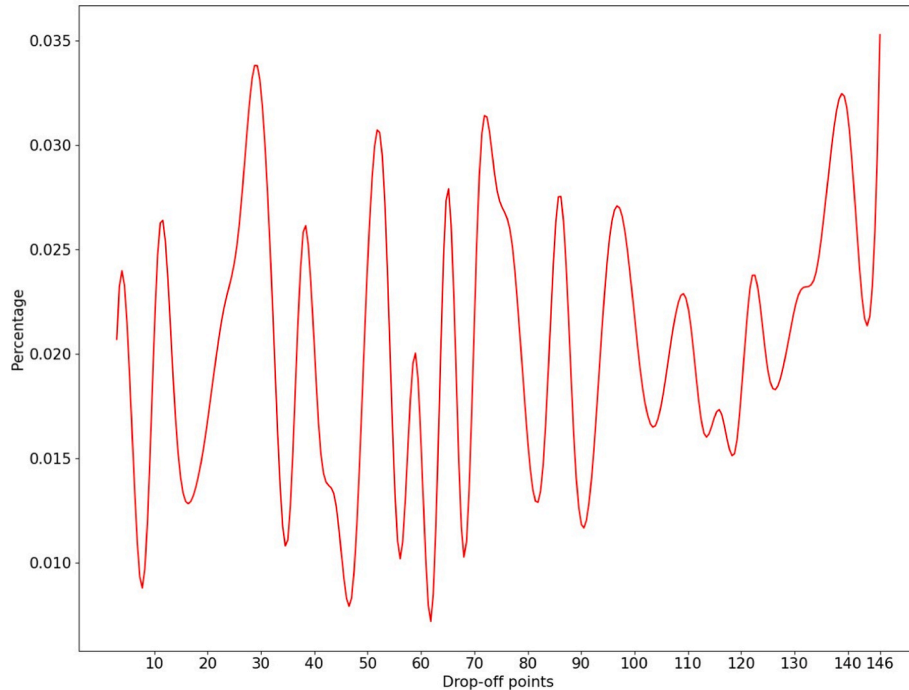


Fig. A3. Multinoulli distribution of drop-off point distributions for arriving parcels

Data availability

Data will be made available on request.

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